# Guide for rescue and recovery workers

Information about rescuing from vehicles of the Volkswagen brand involved in accidents

Version dated: 11/2025

**Legal notice:**
This guide was only created for rescue workers who have undertaken special training in the area of technical support after traffic accidents and are therefore able to carry out the activities described in this guide.
Not all of the vehicles shown in the figures are available in North America.
Specifications and special equipment for Volkswagen vehicles and the vehicles range of Volkswagen AG are subject to ongoing changes.
Volkswagen therefore expressly reserves the right to change these guidelines at any time.
The information takes into account the state of knowledge at the time of creation of this document.
The requirements of the Federal Motor Vehicle Safety Standard, FMVSS No. 305a and the National Highway Traffic Safety Administration (NHTSA) must be observed.

**Please note:**
The information included in this guide is neither intended for end customers nor for workshops or retailers.
End customers can information about the functions of their vehicle and important safety information about vehicle and occupant safety in the vehicle map of the respective Volkswagen AG vehicle.
Workshops and retailers can obtain information about repairs from sources known to them.





Version dated: 11/2025

# Contents

# List of abbreviations

| | |
|---|---|
| AC | Alternating Current |
| AGM | Absorbent Glass Mat |
| BEV | Battery Electric Vehicle |
| $CO_2$ | Carbon dioxide |
| DC | Direct Current |
| EV | Electric Vehicle, only powered by an electric motor |
| GTI | Volkswagen models with a combustion engine (gasoline) |
| HOLD | Prevents a vehicle from rolling unintentionally |
| ID. family | Volkswagen models with a high-voltage drive |
| ID. | Volkswagen models with a high-voltage drive |
| ISO | International Organization for Standardization |
| KESSY | Locking, unlocking and starting without actively using a key |
| Li-Ion | Lithium-ion |
| R134 a | Refrigerant for A/C systems (tetrafluoroethane) |
| R744 | Refrigerant for A/C systems (carbon dioxide) |
| R1234 yf | Refrigerant for A/C systems (tetrafluoropropene) |
| SRS | Supplemental Restraint System |

# Preface

Vehicle and its surroundings: These factors whose interplay are decisive for road safety.
In an accident situation, the vehicle has the following tasks, among others:

- to largely ensure a survival space with a rigid passenger cell,
- to break down the force of impact using intelligent structural concepts and elements,
- using an optimized – consisting of airbags and safety belts with seat belt tensioners and belt force limits, to effectively protect the occupants,
- to minimize hazards from equipment or drive components through the use of safety equipment.

Volkswagen vehicles have proven in international tests that you are among the safest of vehicles. Therefore, accidents and associated injuries cannot be excluded. It therefore remains indispensable to have a short, fast and effective rescue chain.
Designs and equipment offered by Volkswagen itself are taken into consideration. Retrofit solutions and modifications are not taken into account.
This guide has been created in accordance with ISO 17840-3 and is intended to support rescue and recovery workers in carrying out their tasks with the necessary information about the technology of Volkswagen vehicles.
Technical innovations, such as new materials or new drive technologies, make it necessary to adopt an adapted approach performing rescue operations from vehicles involved in accidents.
The processes and approaches are usually governed by service regulations or legislative guidelines or the rescue organization in different countries around the world. If information about the approach is given in these rescue guidelines, these must therefore only be considered as suggestions.
The information are particularly intended for the education and training of rescue and recovery workers. Corresponding rescue data sheets are available for work at the deployment site for Volkswagen vehicles.

**The document contains cross-references and hyperlinks to external sources. Links to external sources are without guarantee.**

**Click on the light blue texts to follow the cross-references. To go back to the last view, click the ↩ button at the top left.**

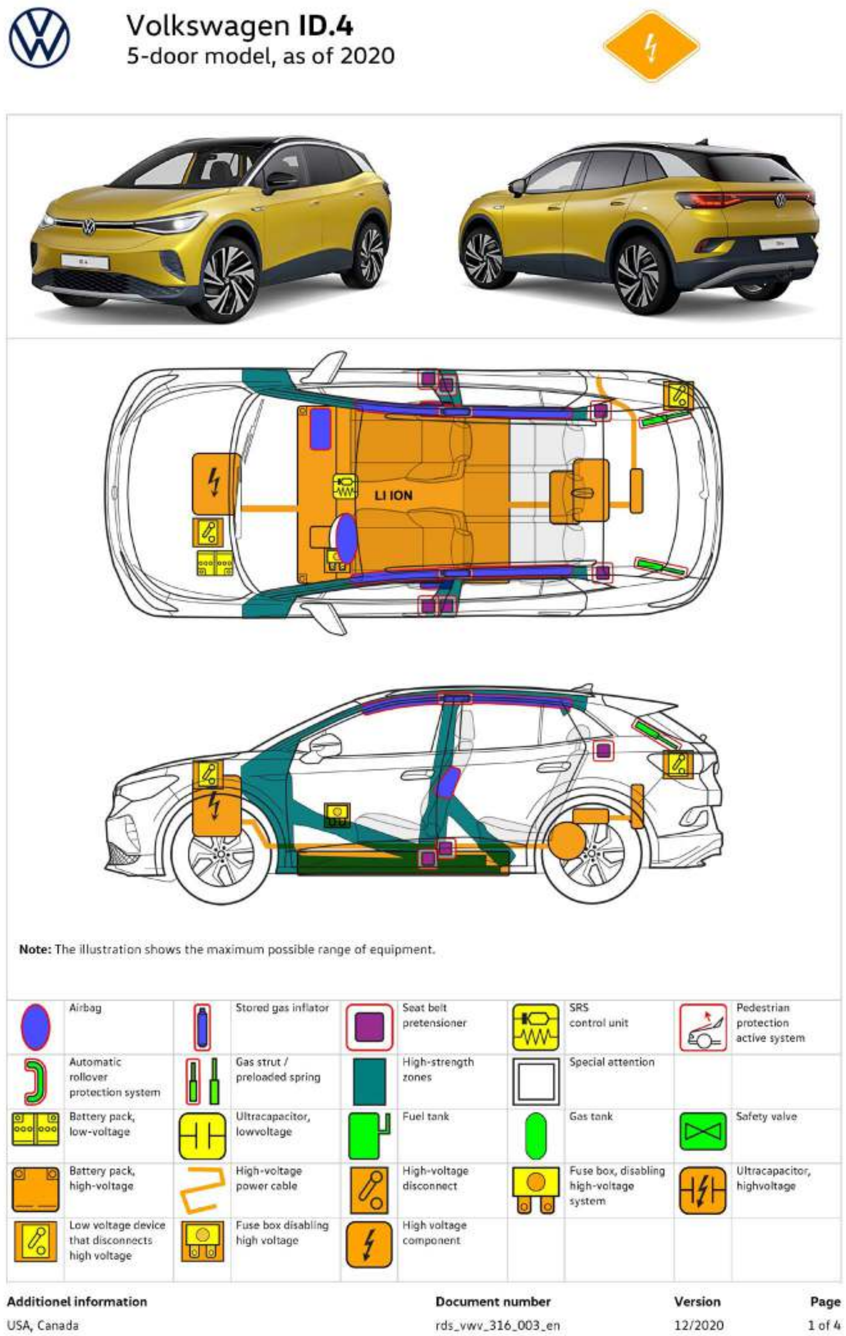
You can find the relevant latest version at www.vw.com.

# 0. Rescue card(s)

# Volkswagen and Volkswagen commercial vehicles provide rescue cards for all models and vehicle versions.

All of the models of the Volkswagen brand are listen in a model overview. The individual rescue cards can be downloaded directly from the model overview.
As an example, the adjacent figure shows the page from the rescue card for Volkswagen ID.4 according to ISO 17840-1:2015.
The current Volkswagen rescue cards are also available at www.vw.com:

**The rescue cards of all Volkswagen models since 2020 were created according to ISO 17840-1 in its relevant applicable form and may differ from each other. For vehicles prior to 2020, the rescue cards are designed according to the manufacturer layout.**



# Area of application

This guide for rescue and recovery workers applies to all vehicles of the Volkswagen brand.
The models are equipped with a petrol drive or a high-voltage drive. The range of vehicle models may vary on a country-specific basis.

For example, Volkswagen models are shown on this and the following pages.

The current Volkswagen model range is available at www.vw.com.

## Labeling of drive types

**Vehicle with class 2 liquid fuels**

**Electric vehicle**

**The drive types are described in the model-specific rescue cards.**

## Volkswagen model range

**Atlas**

**Atlas Sport**

**Golf GTI**

**Golf R**

**Jetta**

**Jetta GLI**

**Taos**

**Tiguan**

## Volkswagen model range

**ID.4**

**ID. Buzz**

# 1. Identification/recognition

## Distinguishing features of Volkswagen models

It is key to identify the vehicle model and its drive type after an accident. Depending on the vehicle model or drive type, specific approaches must be considered as part of a rescue and recovery action.

In addition to the Volkswagen logo, the individual models can be identified by the relevant body shape, body size and the individual vehicle design.
In addition, the model name and the technology lettering on the rear of the vehicle can assist with identification. However, this lettering is missing if it is canceled during the purchase or if it is retrospectively removed.
The figures on this page show how to attach the logo and lettering as an example.

### Volkswagen logo

Volkswagen logo in the radiator grille

Volkswagen logo on the rear lid

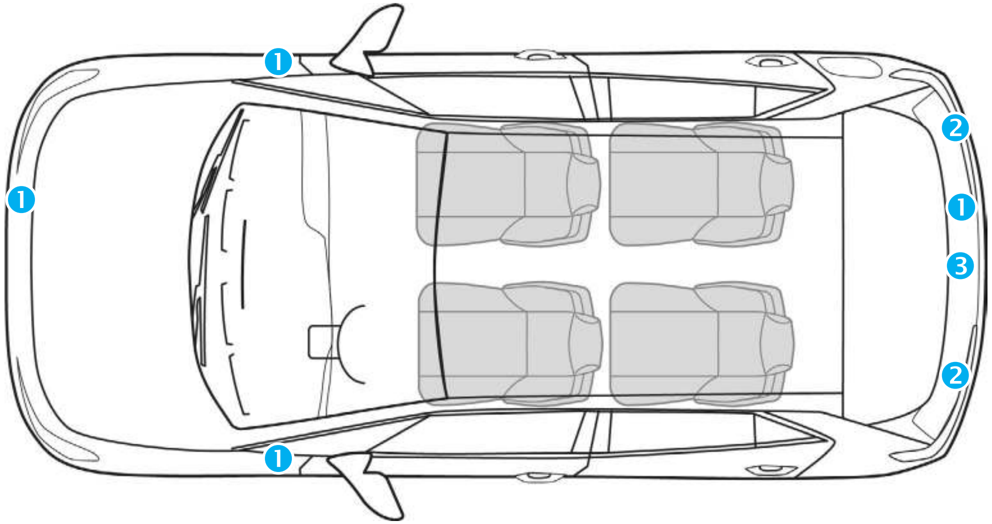
### Model name

Model name on the rear of the vehicle

# Distinguishing features of vehicles with a combustion engine

Volkswagen models with conventional combustion drives (petrol) can be identified using the following features.

**The model-specific distinguishing features are described in the rescue cards.**




## Features on the vehicle

1 Model-specific lettering such as “GTI”, or “R”

2 visible exhaust system

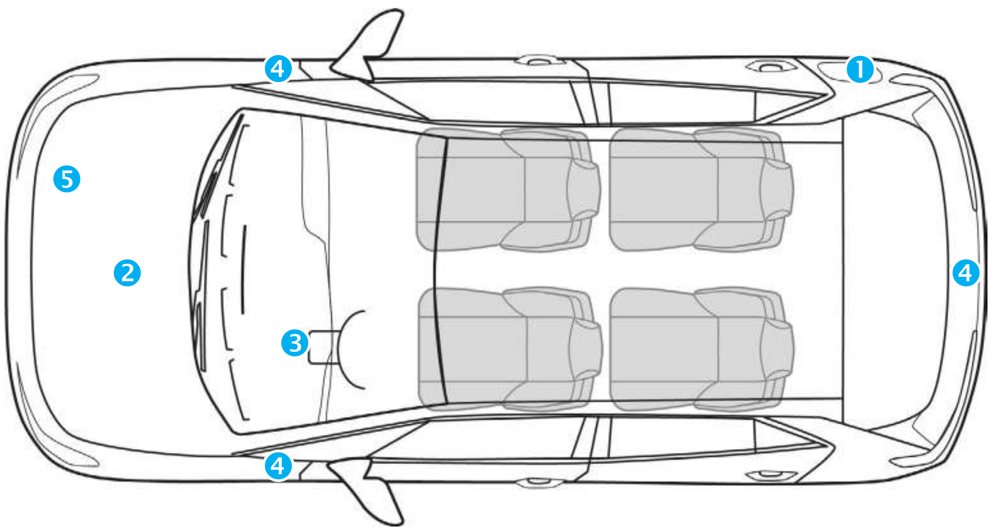
3 Model name centered below the Volkswagen logo

## Distinguishing features of high-voltage vehicles

Volkswagen models with a high-voltage drive are offered with an electric drivetrain (BEV).

**The electric drive motor is silent. The display in the instrument cluster (power display) indicates whether the electric drivetrain is switched off (“OFF”) or is ready for operation (“READY”).**

**The model-specific distinguishing features are described in the rescue cards.**




### Features on the vehicle

1 Charging sockets in the rear fenders

2 Orange color in the vehicle front end

3 The specific displays in the instrument cluster, such as charging indicators, “READY” power meter for drive-ready mode,

4 lettering “ID.” on the rear lid and at the sides of the fenders (lettering may be missing)

5 Warning sticker in front end

## Classification of electrification variants

Following an accident, electric vehicles pose different hazards to rescue and recovery workers to those posed by vehicles with a conventional drive. It is therefore important to identify these vehicles at an early stage in the operation.

Volkswagen offer different electrification variants, which differ with respect to their primary energy source, voltage, type of driven machine and electric range.

A distinction is drawn between the following variants without an external charging socket:

- Mild-Hybrid Electric Vehicle (MHEV)
- Full-Hybrid Electric Vehicle (HEV)

and the following variants with an external charging socket:

- Plug-In Hybrid Electric Vehicle (PHEV)
- Battery Electric Vehicle (BEV)

The table shows the different electrification concepts. Mild-hybrid vehicles (MHEV) with electrical system voltages of up to 48 V are not high-voltage vehicles. Nor does the external appearance of these vehicles differ from conventional Volkswagen vehicles of the relevant model.
All other listed variants are high-voltage vehicles.

| Legend for energy sources | |
|---|---|
| | Traditional fuels, such as gasoline |
| | Battery operation |
| | Battery operation with charging option via socket |

| | Mild-Hybrid (MHEV) | Full-Hybrid (HEV) | Plug-In Hybrid (PHEV) | Battery Electric Vehicle (BEV) |
|---|---|---|---|---|
| Voltage | 12–48 V | 200–300 V | 300–450 V | 300–950 V |
| Electric drive motor | 10–15 kW | 20–50 kW | 60–120 kW | > 150 kW |
| Electric range | | Approx. 1.9 miles | Approx. 31–62 miles | > 124 miles |
| Power source | | | | |
| Models (examples) | | Jetta hybrid<br>Touareg Hybrid | | ID. 4<br>ID. Buzz |

# 2. Immobilization/stabilization/lifting

## The stabilization or fixing of a vehicle reduces the hazards that may result from unintended vehicle movements after an accident.

Contemporary vehicle systems such as the “start-stop system”, Auto Hold (HOLD button) or the new noiseless drive systems give the impression that the vehicle is switched off.
However, depending on the accident situation, these systems may result in the vehicle starting or rolling away unintentionally.
Before starting the rescue operation, it is therefore recommended to ensure that the ignition state is “OFF” or the power display is “OFF” and therefore to disable drive-ready mode. Information about this is given in chapter 3. Disable direct hazards/safety regulations.

In addition, depending on the situation, it is recommended to secure the vehicle against rolling away unintentionally (rolling, tipping, sliding) with wheel chocks, suitable supports or by attaching slings.

**For all current Volkswagen models, the drive-ready mode is automatically disabled with airbag deployment after an accident is detected!**

**For high-voltage vehicles, it is recommended to always locate an accessible high-voltage coupling point in order to de-energize the high-voltage system! Also refer to the chapter 3. Disable direct hazards/safety regulations.**

**The recommended approach for deactivating the high-voltage coupling points is described in the model-specific rescue cards.**

**If the 12 V vehicle battery, all vehicle electrical system functions will be disable (particularly applies to the emergency flashers, interior lighting and electric seat adjustment).**
**For further information, refer to chapter 4. Access to the occupants and chapter 9. Important additional information.**

# Securing the vehicle against rolling away

Volkswagen models may be equipped with a manual transmission or an automatic transmission.
Depending on the accident situation, to prevent the vehicle from rolling away or starting up unintentionally, first move the gearshift lever to “Neutral” (for a manual transmission) or the “P” position for a manual transmission.

1. Select the right/a suitable gear
2. Locate the parking brake
3. Apply the parking brake

**If necessary, secure the vehicle against rolling away unintentionally with suitable wheel chocks or straps.**

Applying the electric parking brake on the right-hand steering column lever.

If additional securing methods are required, the following vehicle areas can also be used: Vehicle columns, carriers, wheels, axles, towing loops or, optionally, the ball hitch.

# Lifting the vehicle

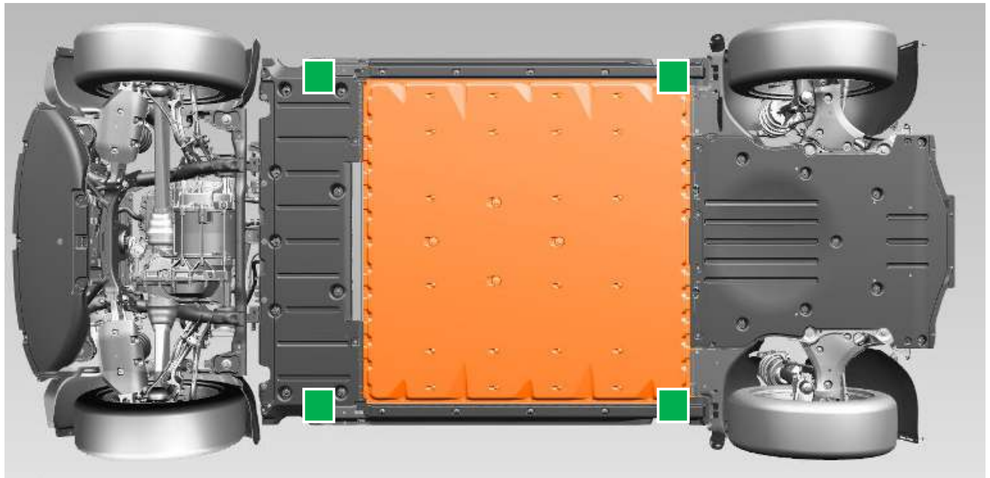
It may be necessary to lift the vehicle in order to rescue injured persons. Ensure that delicate parts, such as the high-voltage battery, drive train, fuel tank or exhaust system are undamaged where possible.

**When lifting or securing vehicles involved in accidents, use firmly fixed components. Do not use high-voltage components or exhaust systems.**

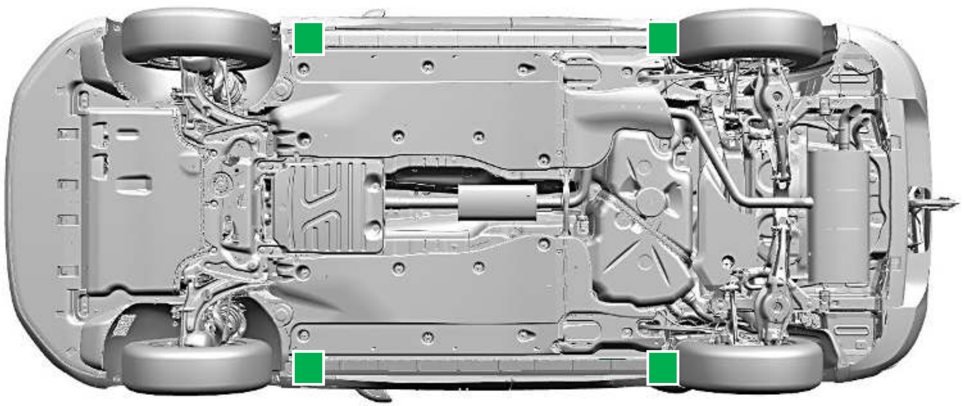
**If vehicles are deformed, the rescue and recovery workers on-site will decide by which points the vehicle can be lifted.**

**Vehicle-specific points for lifting are marked in the rescue cards.**

**If possible, lift the vehicle by the marked lifting points.**



Suitable lifting points on vehicles with a high-voltage drive, e.g. ID.4



Suitable lifting points on vehicles with a combustion engine drive, e.g. Tiguan

Suitable lifting points    High-voltage battery

# 3. Disable direct hazards/safety regulations

In dangerous situations, detecting and redressing dangers to life and limb plays a key role. This chapter describes the suitable preventive measures that reduce the dangers to people involved in accidents and rescue workers.

**Wear suitable protective clothing because liquids or gases could emerge that may lead to injuries or explosions.**
**As part of rescue and recovery operations, you should avoid contact with these substances where possible.**

In dangerous situations, the following approach is recommended:

1. Warn the surrounding area of dangers
   (switch on warning flashers, is automatically activated after an accident)
2. Immobilize the vehicle, see Chapter 2. Immobilization/stabilization/lifting
3. Disable direct hazards
   Switching off the ignition
4. De-energize the vehicle electrical systems
   Disabling the high-voltage system,
   Disconnecting the 12 V vehicle battery (depending on the situation),

**In accidents with airbag deployment, the high-voltage system and the 48V electrical system are automatically deactivated.**
**The high-voltage system will be de-energized around 20 seconds after deactivation.**

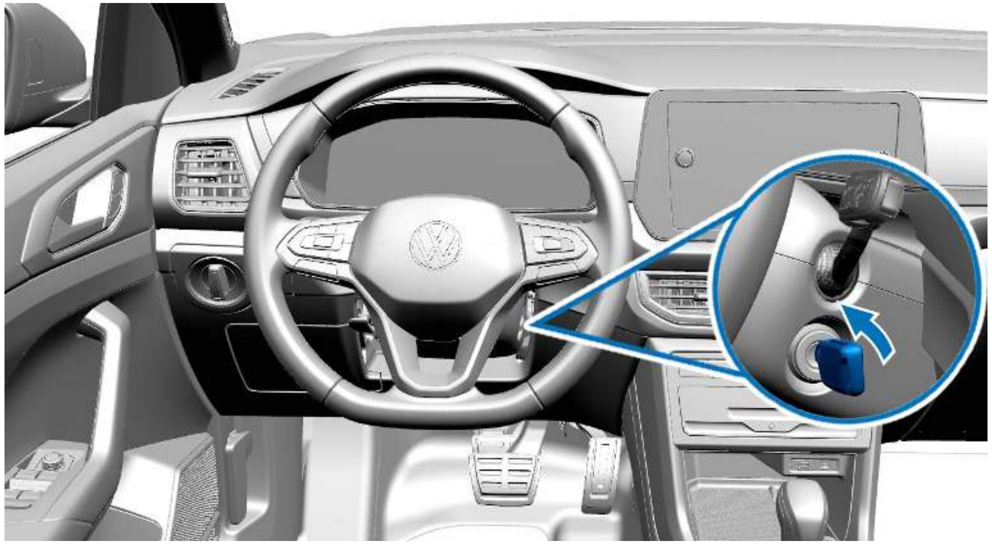
# Switching off the ignition

Turn the remote control vehicle key to “OFF” and remove it. Many Volkswagen models are equipped with a “START-ENGINE-STOP” button. This can be found on the steering column, in the center armrest or in the instrument panel.

Take the following options, among others, into account:

- The vehicle has either a conventional ignition lock or KESSY, a system in which the ignition lock can be located anywhere inside the care in order to enable the vehicle (e.g. in the driver’s trouser pocket or a handbag in the vehicle). In addition, there is the option of managing the vehicle with an app.
- Use the remote control vehicle key, where available, to set the vehicle to “OFF”.

If the vehicle has a “START-ENGINE-STOP” button with which the vehicle can be disabled, press it.
Then remove the remote control vehicle key, the valet keycard or the smartphone from the vehicle and keep it a minimum of 5 m away from the vehicle in order to prevent unintentional activation.



**Vehicle with a conventional key.**

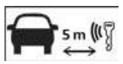
**Vehicle with a “START-ENGINE-STOP” button in the center console**

## 3. Disable direct hazards/safety regulations

Vehicle with a “START-ENGINE-STOP” button on the steering column

**If you simultaneously press the “START-ENGINE-STOP” button and apply the brake pedal, the vehicles may switch to drive-ready mode!**
**Refer to the information in the rescue cards!**

**For vehicles with a high-voltage drive, the “Powermeter” display in the instrument cluster reports whether the electric drivetrain is “OFF” or “READY”.**



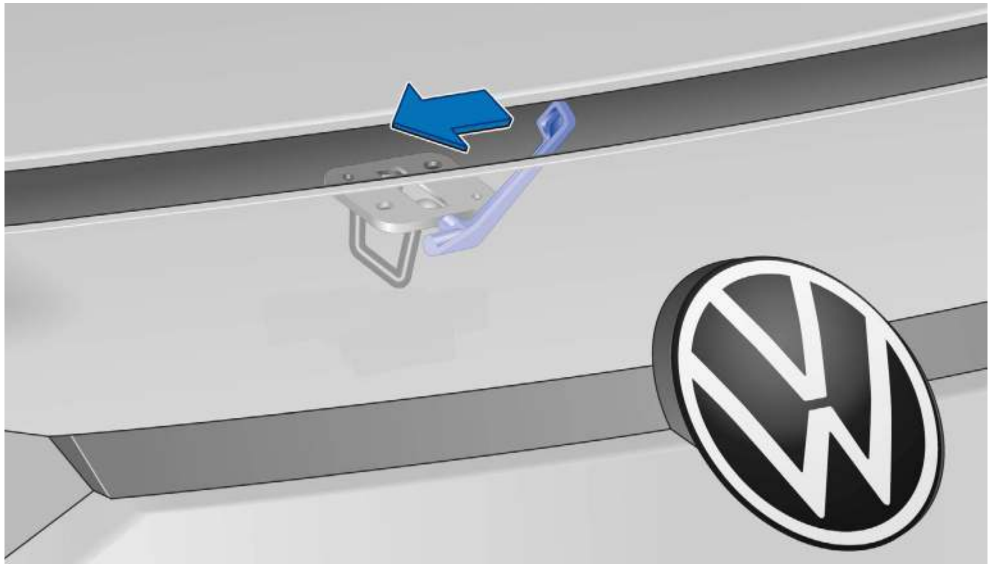

**It is also possible to use a digital key card or smartphone app instead of a remote control vehicle key.**
**Remove the remote control vehicle key, digital key card or smartphone from the vehicle (while maintaining a minimum distance of 15 ft from the vehicle).**

# Opening and closing the hood

Depending on the situation, it may be necessary to open and close the hood.
The following section describes the standard procedure (the 2-lock system is not taken into account).

**Further information is described in the model-specific Owner's Manual.**

In the footwell on the driver side: Release lever for the hood.



On the hood: Operating lever.

## Disabling the high-voltage system

### Volkswagen models with a battery electric drive (BEV) are equipped with a high-voltage system with a voltage of above 300 V.

The high-voltage system is immediately separated from the high-voltage battery after the airbag deployment is detected. The high-voltage system can then only be activated by a suitable qualified workshop. In addition, indicators or warnings can be displayed on the instrument panel.
Volkswagen high-voltage vehicles have multiple emergency cut-out connections, which are located on the fuse box, in the front end or on the rear of the vehicle. They offer rescue workers a good opportunity to deactivate the high-voltage system safely. You can find more information on the following pages at Disconnecting the high-voltage system from the vehicle.

**The electric drive motor in vehicles with a high-voltage drive is noiseless. The display in the instrument cluster (power display) indicates whether the electric drivetrain is switched off ("OFF") or is ready for operation ("READY").**

**In vehicles from the ID. Family, the drive-ready mode can be established by the driver seat being occupied and stepping on the brake pedal!**

**If there are accidents with airbag deployment, the high-voltage system will be automatically deactivated. The high-voltage system will be de-energized around 20 seconds after deactivation.**

**For all other cases, an emergency cut-out connection can be used to disable the high-voltage system.**
**In particular, the use of the emergency cut-out connection prevents unintentional reactivation.**

Depending on the accident and situation at the accident location, it is possible that the prioritized emergency cut-out connection on the fuse box may not be accessible. If necessary, the alternative emergency cut-out connections in the front end or the vehicle rear can be used.

There emergency cut-out connections marked with yellow flags only carry the 12 V power supply and can therefore be safely disconnected by rescue workers in accordance with the approach described on the little flags.

**Disconnecting a marked emergency cut-out connection only disables the high-voltage system.**
**Safety systems such as airbags or seat belt pretensioners are still supplied with voltage by the 12V vehicle electrical system.**

**If the airbag is not triggered, after the vehicle electrical system voltage is disconnected, the 12V electrical equipment is still supplied with electrical power via the DC converter from the high-voltage battery.**

**Even after the high-voltage battery is disabled, there is still power present inside the high-voltage battery.**
**During the rescue measures, the high-voltage battery must be neither be damaged nor opened.**

**Do not touch damaged high-voltage components; cover with suitable aids if necessary!**
**Wear personal protective equipment in accordance with local standards!**

**The location of the emergency cut-out connections and the approach for deactivating the vehicle are specified on the Volkswagen rescue cards.**

## At the accident site

Depending on the accident situation, restraint systems, e.g. airbags may be triggered. The rescue and recovery workers at the accident site decide about how to proceed with the rescue and recovery.

**A rapid or heavy smoke development at the accident vehicle may indicate a thermal reaction of the high-voltage battery, see also Fire in high-voltage vehicles.**

### Minor accident

First, no damage is visible and the restraint systems were not triggered.

1. Warn the surrounding area of dangers:
   Switch on warning flashers, set up warning triangle
2. Immobilize the vehicle:
   Chapter 2. Immobilization/stabilization/lifting
3. Disable the high-voltage system:
   By pulling the fuse on the fuse panel or separating alternative emergency cut-out connections

### Severe accident

The restraint systems were triggered. There is initially no visible signs of damage to the high-voltage battery.

1. Warn the surrounding area of dangers:
   Switch on warning flashers, set up warning triangle
2. Immobilize the vehicle:
   Chapter 2. Immobilization/stabilization/lifting
3. The high-voltage system was automatically disabled

**Damage or deformation of the high-voltage battery at the accident vehicle may indicate a thermal reaction of the high-voltage battery, see also Fire in high-voltage vehicles.**

**Depending on the accident situation, it is also necessary to manually disable the high-voltage system at an emergency cut-out connection.**

### Parked or stationary vehicle

If a parked vehicle is damaged by an accident, restraint systems or airbags are not usually triggered. The high-voltage system is not automatically disabled. If the ignition is switched off, warnings cannot be displayed on the instrument panel.

1. Disable the high-voltage system by pulling the fuse on the fuse panel

### Vehicle at charging station

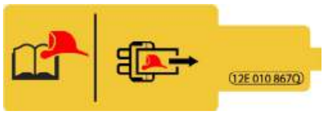
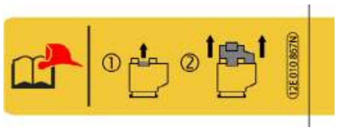
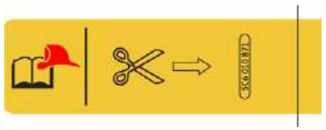
If a charging vehicle is damaged by an accident, restraint systems or airbags are not usually triggered. The high-voltage system is not automatically disabled. If the ignition is switched off, warnings cannot be displayed on the instrument panel.

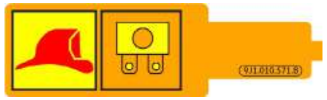
1. Regularly remove the charging cable (see vehicle’s Owner's Manual)
2. Alternative Disconnecting from the charging station (emergency release)

**The high-voltage components are indicated by warning signs, see also Warning labels on high-voltage components. High-voltage cables are orange in color**

## Labeling of emergency cut-out connections

The emergency cut-out connections for disabling the high-voltage system are uniformly labeled in Volkswagen Group models. The pictograms on the labels explain how to proceed.
Until 2022, the labels were manufactured according to individual specifications and fitted to the models. Currently, new labels aligned with the EURO NCAP are used. These labels will also be used by all Volkswagen Group models in the future,

| Previous labeling | |
|---|---|
|  | **Labeling of emergency cut-out connection in the passenger compartment (pull out fuse on fuse panel)** |
|  | **Labeling of emergency cut-out connection in the front end (open the maintenance connector)** |
|  | **Labeling of emergency cut-out connection in luggage compartment or at the rear of the vehicle (cut through identified cable)** |

| New labeling from 2023 | |
|---|---|
|  | **Labeling of emergency cut-out connection in the passenger compartment (pull out fuse on fuse panel)** |
|  | **Labeling of emergency cut-out connection in the front end (open the maintenance connector)** |
|  | **Labeling of emergency cut-out connection in luggage compartment or at the rear of the vehicle (cut through identified cable)** |

## Disconnecting the high-voltage system from the vehicle

**The electric drive motor in vehicles with a high-voltage drive is noiseless. The display in the instrument cluster (power display) indicates whether the electric drivetrain is switched off ("OFF") or is ready for operation ("READY").**
**Refer to the information on the relevant rescue cards.**

If the high-voltage system also needs to be separated manually, follow the following sequence:

1. First, use the Main method: High-voltage coupling point on fuse panel; if this cannot be achieved, use
2. Alternative method: High-voltage coupling point in the vehicle front end (maintenance connector for high-voltage system) or
3. Alternative method: High-voltage coupling point in the rear of the vehicle .

There is a minimum of two coupling points in the current Volkswagen models. One is located in the fuse panel and another is fitted in the front end. In some vehicles of the ID. family, there is also a third coupling point in the rear of the vehicle.

Depending on the vehicle type and equipment, different approaches may be offered. The procedure for disabling is based on the accident situation and vehicle equipment.

**The assembly point of emergency cut-out connections and the required approaches can be found in the Volkswagen rescue cards.**

**There is only the greatest possible certainty that the high-voltage system is disabled when an emergency cut-out connection provided by a manufacturer is separated and the 12 V on-board battery is disconnected.**

### Use rescue equipment with car and caution in the vicinity of high-voltage components

Irrespectively of whether it is a hybrid or electric vehicle, the following points are generally valid during rescue operations on high-voltage vehicles.

**If high-voltage components are handled incorrectly, there is a risk to life due to the high voltage and the possible current flow through the human body.**

**It is not admissible for any work operations to be performed on very badly damaged high-voltage components. One of the accessible emergency cut-out connections should also be opened.**
**If the airbags have not triggered, the vehicle must be disabled by the rescue and recovery workers via an emergency cut-out connection. After approx. 20 seconds, the high-voltage system will be de-energized.**
**If the airbags have triggered, the high-voltage system has already been switched off; there is no further waiting time for the rescue and recovery workers.**

**Even after the high-voltage battery is disabled, there is still electric power present inside the high-voltage battery. During the rescue measures, the high-voltage battery must be neither be damaged nor opened.**
**If the high-voltage battery has been damaged by the effects of the accident, avoid contact with the high-voltage battery or with liquids or vapors leaking from the high-voltage battery.**

**Do not touch damaged high-voltage components; cover with suitable aids if necessary!**
**Wear personal protective equipment in accordance with local standards!**

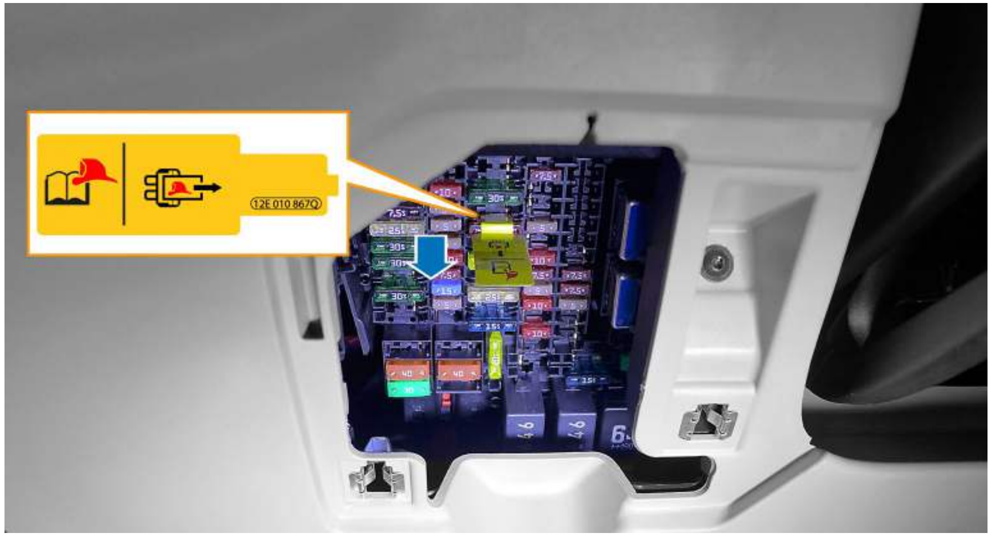
| | |
|---|---|
|  | **New labeling for high-voltage disconnection on fuse panel** |
|  | **Fuse for switching off the high voltage** |

**The location of the emergency cut-out connections and the approach for deactivating the vehicle are specified on the Volkswagen rescue cards.**

## Main method: High-voltage coupling point on fuse panel

Depending on the vehicle model, the fuse panel is attached to the vehicle interior in the area of the instrument panel or the vehicle rear and is marked with a yellow flag. The high-voltage system is separated, and therefore disabled, by pulling the fuse identified in this way from its holder.

The high-voltage contactors in the high-voltage battery are opened and are disconnected from the remaining high-voltage system, which is then de-energized after 20 seconds have elapsed.



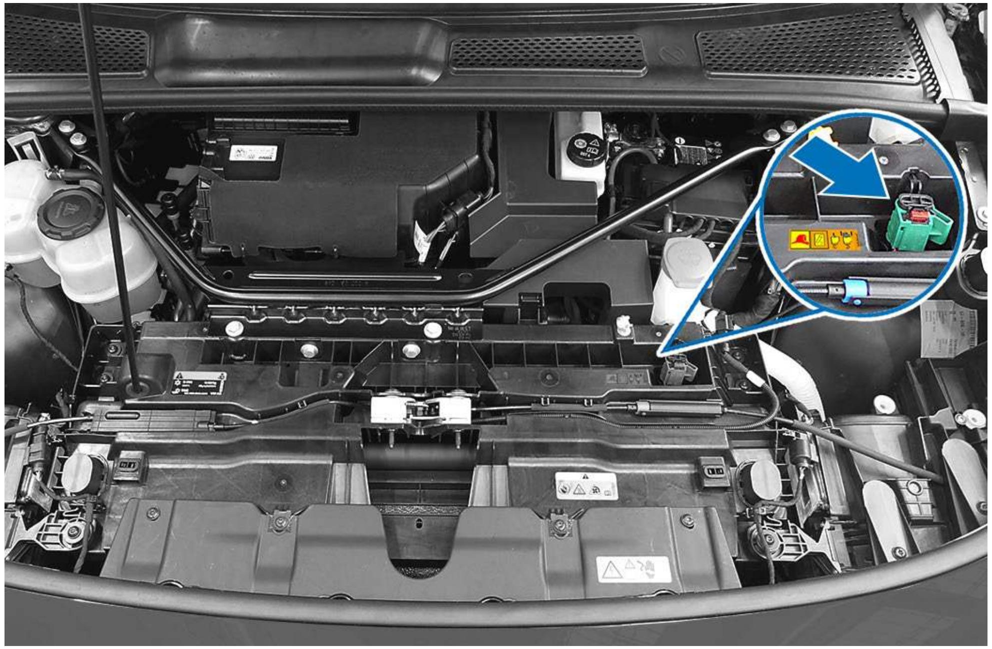

**Coupling point in passenger compartment, instrument panel on fuse panel.**

## Alternative method: High-voltage coupling point in the vehicle front end

The low-voltage maintenance connector for high-voltage system in the vehicle front end is used in electric vehicles (BEV) acts as an emergency cut-out connection for the high-voltage system. The connector has a green connector housing and a tab for unlocking. A yellow label on the connector cable clearly indicates that the connector is an emergency cut-out connection. It can then only be activated by a suitable qualified workshop.
Access to the vehicle front end is usually enabled by pulling the release cable in the front left footwell. This releases the front compartment lid, which can then be installed. If necessary, please refer to the Owner's Manual for the vehicle.

| | |
|---|---|
|  | **New labeling for high-voltage disconnection in engine compartment** |
|  | **High-voltage disconnection at low-voltage coupling point** |

**If the label of the emergency cut-out connection in the vehicle front end is not visible, an additional sticker can be attached in the vicinity,**



**Coupling point in electric vehicle front end**

Procedure for disabling the high-voltage system using the emergency cut-out connection:

**Pull out red tab**

**Press and hold the red tab and at the same time pull out the black connector until it locks in position.**

## Alternative method: High-voltage coupling point in the rear of the vehicle

In vehicles from the ID. Family, an additional coupling point can be found in the rear area. This is where a cable, marked by a small yellow flag, must be cut through.


**Alternatively, the trim can be unclipped under the taillight and the coupling point can be cut through.**

 **Cable coupling point**

## Disconnecting from the charging station (emergency release)

Vehicles parked at a charging station or a home charging station for charging can be disconnected from these in an emergency.

If regular separation is not possible, the vehicle can be released using an operation described in the rescue card. In principle, the emergency release is always located on the back of the charging socket.

**The procedure for using the emergency release of the charging connector on the vehicle side is described in the rescue cards.**

**If necessary, public charging stations for supplying power are connected to the public grid with over 1000 V of voltage. If this is the case, the relevant safety distances must be complied with in the event of a fire.**

**Comply with regional and country-specific operational plans and safety information for rescue and recovery workers for public charging stations and home charging stations!**

**The charging sockets and appearance of public and private charging stations vary depending on the manufacturer and country.**

Charging Stations and home charging stations charge with alternating voltage or direct voltage.
For a system with direct voltage (DC), the battery is directly provided with power via the charging socket.
If alternating voltage (AC) is used to charge the high-voltage battery, the battery charger in the vehicle assumes the function of the DC/DC converter.

## Disconnecting the 12 V vehicle battery

The situations at accident location may require the 12V vehicle electrical system to be disabled in order to reduce hazards toward accident victims or rescue workers (e.g. retrospectively triggering airbags).

Depending on the vehicle type and equipment, one or more 12 V vehicle batteries may be fitted.
By disabling the vehicle electrical system, first, the risk of fire due to short circuits, and also, the risk of retrospective activation of airbags, seat belt pretensioners or the rollover bar is reduced.
When the vehicle electrical system is disabled, it must be ensured that the power supply is disconnected from any trailers and any solar elements in the sliding sunroof is covered.

**After disconnecting the 12-volt vehicle electrical system voltage, the airbags will be deactivated. In the event of fire, airbags that have not ignited may deploy due to heat!**

**After disconnecting the 12-volt vehicle electrical system voltage, electrically operated trailer hitches can no longer be used!**
**Refer to section Recovering vehicles involved in accidents.**

**If multiple on-board batteries are fitted, they must all be disconnected so that the vehicle is de-energized.**

**If vehicles in which access to the on-board battery is not possible: Volkswagen has an accessible earth cable fitted from the battery to the vehicle body; disconnect this. After disconnecting the earth cable, always insulate this cable in order to reduce the risk of electric arcs.**

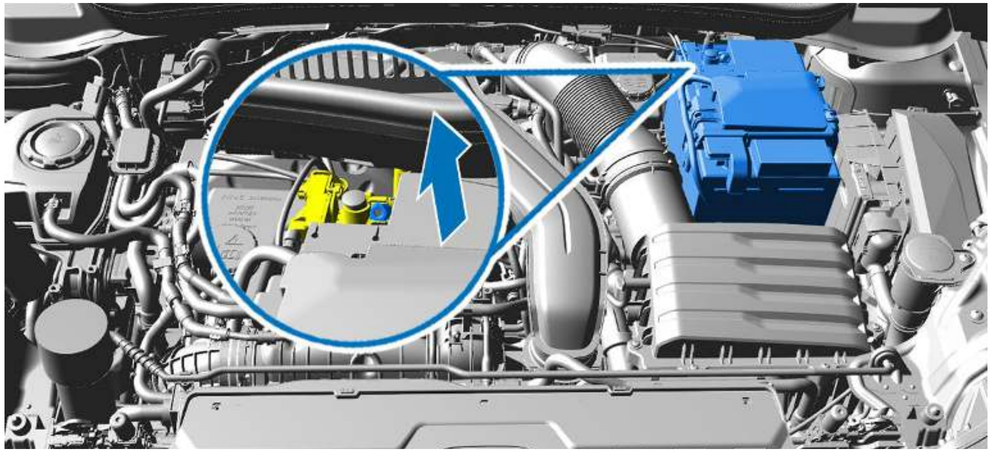
**Always first disconnect the negative battery terminal and then the positive battery terminal from the battery. In order to avoid the risk of electric arcs, the battery terminals should be insulated.**

**If the 12 V supply is disconnected, all vehicle electrical system functions will be disable (particularly applies to the emergency flashers and electric seat adjustment).**
**For further information, refer to chapter 4. Access to the occupants and chapter 9. Important additional information.**

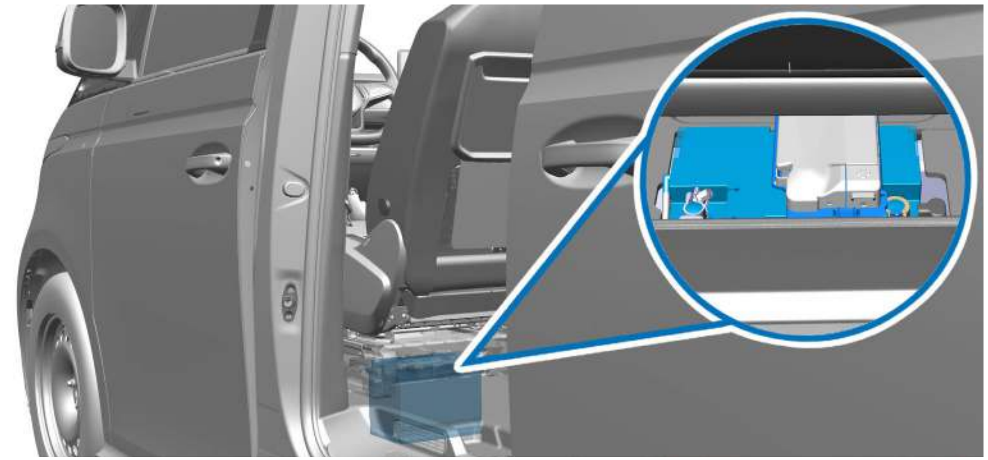
**The installation location and the required approach for disabling the 12 V vehicle electrical voltage is described in the Volkswagen rescue cards.**

## Typical fitting locations

Depending on the requirement, the 12 V on-board battery is located in the front end, luggage compartment or passenger compartment. A second 12 V vehicle battery may also be fitted at another location in the vehicle. In some Volkswagen Commercial Vehicle models with camping equipment, additional 12 V vehicle batteries may be fitted.



**Installation location in the front end**



**Installation location in passenger compartment below driver seat.**

# 4. Access to the occupants

# As part of rescue activities after an accident, access to vehicle occupants plays a central role.

Depending on the accident situation, there are various access options to the vehicle occupants for rescue and recovery workers.

## Releasing vehicle doors

Locked doors (exterior door handle not functioning) can be regularly unlocked as follows

- Buttons on the remote control
- Button on the door trim panel
- Manual remote control vehicle key/optional keyless
- Optionally via app/keyless card

**Buttons on the remote control of the remote control vehicle key**

**Button on the door trim panel**

**Vehicle-specific or equipment-specific information can be found in the on-board literature or the model-specific rescue cards.**

**After an accident with deployment of airbags, the vehicles doors and rear lid are automatically unlocked. The doors can be opened by strongly pulling the exterior door handle.**

## Electrically assisted door handles

For some Volkswagen models, activating the door handles internally and externally are electrically assisted. The doors can be conveniently unlocked and opened with very little effort.
In the event of a crash, a considerable amount of force may be required to unlock and open the doors.

**In the event of an accident or if the 12V vehicle electrical system fails, a considerable amount of force is required to open the doors.**

**In the event of accidents where the airbag is deployed, all doors and hoods are unlocked automatically.**

**After serious accidents, it may also be necessary to use tools.**

**Where possible, the electrical convenience equipment should be used before disconnecting the battery for the benefit of rescue operations.**

## Exterior door handles

Electrically assisted door handles enable all doors to be unlocked and opened with little force. To open, grip in the recessed grip and gently fold up the door handle.
If the electrical assistances is interrupted or fails, the door handle must be levered upward with a higher amount of force.

1. **Convenience opening: Gently lift the door handle and open the door.**
2. **Emergency opening: Lever the door handle far upward with a higher amount of force and open the door.**
**(Example: Door handle from ID.4).**

In particular situations, the vehicle can be manually unlocked and opened from outside with an emergency key as follows:

1. Pry off the cap clockwise with the remote control vehicle key.
2. Put the key bit into the lock cylinder.
3. In order to unlock the vehicle, turn the emergency key anticlockwise.
4. In order to open the door, pull on the driver's door handle with force.

If necessary, the vehicle doors can also be unlocked and open from the inside by operating the interior door handle.

**If the child safety lock is activated, it is not possible to open the doors for the second row of seats from the inside. In order to open the door from the inside, the child safety lock must first be mechanically or electrically disabled.**

**Mechanically disable the child safety lock at the door using a digital key**

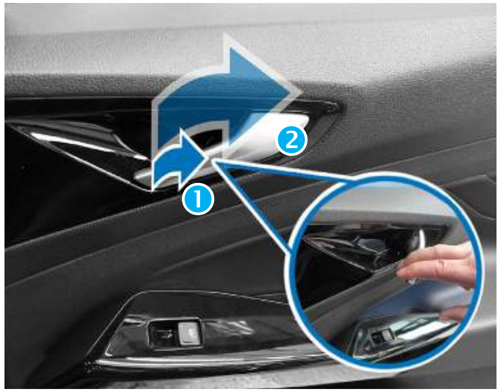
**Electrically disabling the child safety lock in the door trim**

**panel**

## Interior door handles

The doors of electrically assisted door handles can also be operated conveniently from the inside. To do so, gently fold down the interior door handles and open the doors. Even if the electrical assistance is interrupted or fails, the doors can be opened by pulling the door handle. The interior door handles are accessed through:

1. Adjacent door
2. Opening the window via:
   - Buttons on the remote control,
   - Button in the door trim panel
3. Removed window




❶ Convenience opening: Gently fold down the door handle and open the door.

❷ Emergency opening: Lever the door handle far backward with a higher amount of force and open the door. (Example: Door handle from ID.7).

**There is a risk of injury if the vehicle window panes shatter! Use appropriate personal protective equipment!**

**If anti-play protection is enabled, it is not possible to operate the windows in the second row of seats. To open the windows, the anti-play protection is disabled.**

**After accidents where airbags are deployed, the windows move to a crash position (gap of approx. 5 cm). If necessary, the window can be broken through by reaching outward.**

## Access via rear lid

Depending on the equipment variant, the rear lid can be unlocked as follows:

**Button on the rear lid button of the remote control**

**Button in the driver's side door trim panel**

The rear lid is opened by pressing the electrical button in the rear lid when the vehicle is unlocked. Some models have the option of electrically openings the rear lid.

**In the event of accidents where the airbag is deployed, all doors and hoods are unlocked automatically.**

**If the 12V power supply is interrupted, it is not possible to open the rear lid despite unlocking being performed.**

**If necessary, the rear lid can be manually opened from the inside. Please note the information in the model-specific Owner's Manual.**

## Electrically-powered sliding doors

Some vehicles from the Volkswagen Commercial Vehicle brand may be equipped with one or more sliding doors that are electrically operated.
After an accident, electrically-powered sliding doors behave in the same way as usual mechanical doors.

**After an accident or if the 12V vehicle electrical system fails, a considerable amount of force is required to open the doors.**

**In the event of accidents where the airbag is deployed, all doors and hoods are unlocked automatically.**

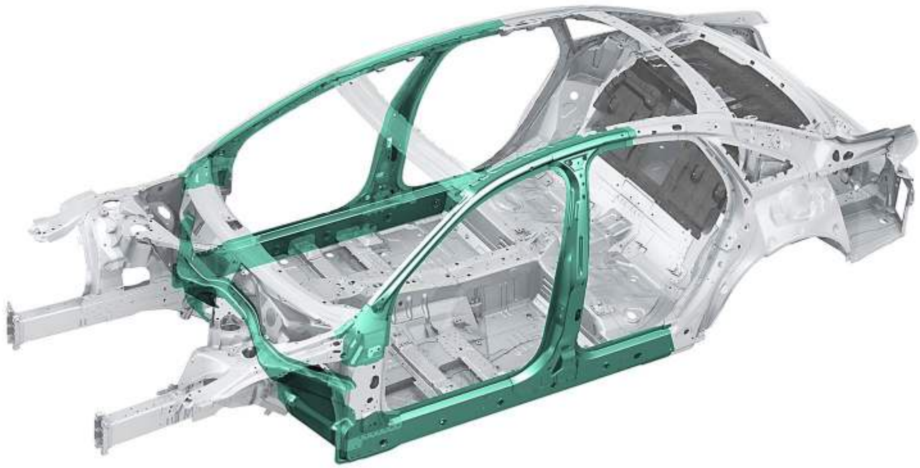
**If the child safety lock is activated, it is not possible to open the doors from the inside. In order to open the door from the inside, the child safety lock must first be mechanically or electrically disabled.**

**Where possible, the electrical convenience equipment should be used before disconnecting the battery for the benefit of rescue operations.**

## Vehicle body reinforcements

In particular, a high degree of safety for vehicle occupants is achieved by a rigid passenger cell.

The rigid nature of the vehicle body is achieved through the use of high-strength and hot-formed steels with thicker walls in a multi-layered construction.



**Vehicle body with rigid passenger cell**

The rigid areas are shown in the model-specific rescue cards. When rescue operations are carried out, powerful tools must be used in these areas.

**Sharp edges can arise when cutting high-strength or hot-formed steels! Use appropriate personal protective equipment!**

**Avoid sensitive components, such as airbags, fuel tanks, lines or high-voltage components. Information about the position of reinforcements can be found in the model-specific rescue cards.**

**High-strength zones**

## The A-pillar

In Cabriolets, also reinforce the vehicle body in order to achieve the relevant stiff nature of the vehicle body even without a roof. For example, pipe reinforcements are integrated in the A-pillar in order to ensure the protected area together the rollover bar in the event of vehicle rollovers. If necessary, it is possible to open the Cabriolet roof (generally a fabric roof) in the conventional way or by pushing up the roof with a rescue ram.

A-pillar reinforcement in the Cabriolet

**It is only possible to cut through the A-pillar in the area of the A-pillar reinforcement using powerful rescue equipment.**

**The location of special reinforcement measures in the individual vehicles can be found in the rescue cards.**

## The B-pillar

The use of high-strength and hot-formed steels and a multi-layered construction with a large cross-section reinforces the B-pillar, in particular.

In the area of the belt guide, the B-pillar is additionally reinforced, which makes cutting through more difficult. These areas should therefore be specifically bypassed.

B-pillar with multi-layered construction

**Cutting through vehicle pillars in the area above the safety belt height adjuster is the easiest!**
**The pillar can also be cut through in the lower area. However, when doing so, ensure that the cross-section of the pillar is very large and the seat belt pretensioner is usually located there.**

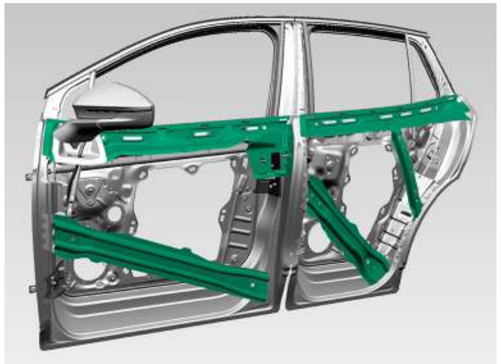
**In any case, the rescue cards must be referred to!**

## The side members

High-strength and multi-layered steels are used in the vehicles to reinforce the side members. These are used to increase safety during a side crash.
In particular, the electric vehicles have reinforced side members in order to protect the high-voltage battery.

## The impact protection in the door area

In Volkswagen Group vehicles, the impact protection in the door area consists of steel pipes or steel profiles. The pipes or profiles are arranged horizontally or diagonally behind the outer door panels.
The high-strength profiles can be separated using powerful cutting equipment. The steel pipe is located above the door lock and offers support for the vehicle in the frontal impact, whereas the steel profiles below the door lock are relevant for side impacts.



Side impact protection in the doors

**A crash pipe may be fitted in the front doors in order to more effectively protect the vehicle occupants during a side impact.**

**The location of special reinforcement measures in the individual vehicles can be found in the rescue cards.**

**High-strength zones**

# Glazing

## The vehicle panes in Volkswagen Group consist of single-pane and laminated safety glass.

The windshield is always made as laminated glass and, depending on the vehicle equipment, the side and rear windows are designed as tempered safety glass or laminated glass. Panoramic glass roofs are used in tempered safety glass or laminated glass by Volkswagen.

### Tempered safety glass

Tempered safety glass is thermally pre-treated glass that can withstand high loads. If the load is too high, it will shatter into many fragments.
Tempered safety glass is used for side windows, rear windows, the sliding sunroof and panoramic roof.

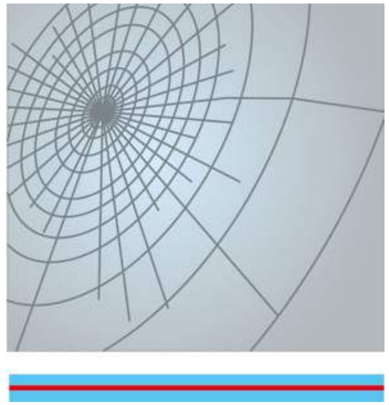
**Intact panes may suddenly shatter during rescue work on the vehicle. Depending on the accident situation and the scope of the rescue work, the panes should be removed beforehand.**
**Panes can be removed by point loading, e.g. with a center punch or an emergency hammer. The panes should therefore be secured by taping them off beforehand.**

### Laminated glass

Laminated glass consists of two glass panels and one intermediate layer of film. The glass panes remain largely intact in the event of damage. They are used for wind-shields and, if necessary, for side windows. The windshields are glued to the vehicle body.

**As the laminated glass cannot suddenly shatter, they must only be removed when it is necessary for the rescue work. Laminated glass panels can be removed with special laminated glass saws or hooligan tools.**


Tempered safety glass


Laminated glass

**Before removing the glass panes, protect the vehicle occupants and rescue works against glass splinters.**

**For newer models, information about the fitted pane variants can also be found in the respective rescue cards.**

# Mechanisms for adjusting the height and length of the driver's seat and steering wheel

Depending on the situation at the accident location, the rescue and recovery workers decide whether it is necessary to adjust the seats or the steering wheel in order to rescue vehicle occupants.

The seating and steering columns in the Volkswagen vehicle models can be operated mechanically or electrically. If necessary, the head restraints must also be removed. In order rescue vehicle occupants from the second and third rows of seats, it may be necessary to move the front seats forward and fold down the backrests or remove the individual seats.

**If rescue tools are used in the vehicle interior, ensure that sensitive parts, such as the high-voltage battery or pyrotechnic seat belt pretensioners are not damaged.**

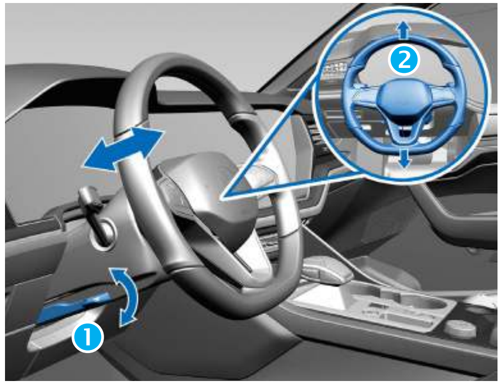
**In the event of accidents where the airbag is deployed, electrically powered doors and the trunk lid are unlocked automatically.**

**Depending on the vehicle equipment, electrically operated seats may be fitted with a convenient entry version. This function automatically moves the seat into various positions.**

**Where possible, the electrical convenience equipment should be used before disconnecting the battery for the benefit of rescue operations.**

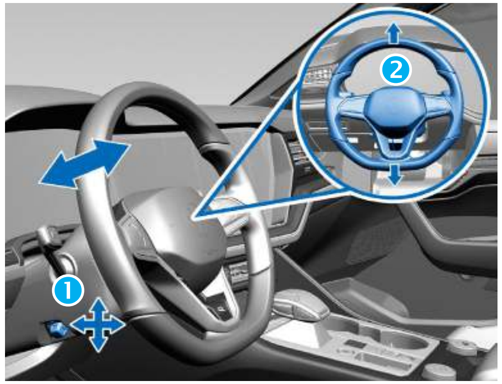
**Mechanical steering column adjustment**

1. **Unlock steering column**
2. **Adjust steering column**




**Electric steering column adjustment**

1. **Unlock steering column**
2. **Adjust steering column**




**Electric seat adjustment**

1. **Adjust lumbar support**
2. **Adjust seat frame**
3. **Adjust backrest frame**

# Electrical convenience equipment

Depending on the model range and vehicle equipment, Volkswagen AG vehicles have a wide range of electrically powered convenience equipment, e.g.:

- Electric doors
- Power windows
- Electric sliding sunroof
- Electric seat adjustment
- Electric steering column adjustment
- Electric releasing, opening and closing of the luggage compartment

After disconnecting the vehicle electrical system battery(ies), these systems can no longer be operated!

| |
|---|
| **In the event of accidents where the airbag is deployed, electrically powered doors and the trunk lid are unlocked automatically.** |
| **Where possible, the electrical convenience equipment should be used before disconnecting the battery for the benefit of rescue operations.** |
| **The battery should only be reconnected to the vehicle electrical system by workshop staff.** |

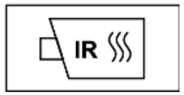
# 5. Stored energy/liquids/gases/solids

Volkswagen models carry a variety of operating equipment. It is only possible to respond appropriately to a hazard and to take suitable measures to avert the hazard if you detect a hazard during use.

**When handling leaking operating equipment, always use appropriate personal protective equipment.**

**List of possible stored energies/ liquids/gases/solids:**

**In the event of mechanical deformation of the battery system, there is the risk of a thermal reaction in the high-voltage battery. Monitor the temperature of the high-voltage battery!**




**For all carried or stored energies (pyrotechnic seat belt pretensioners, airbags, gas-pressure struts, fuels, gases, etc.), there is the risk of an expansive discharge after an accident.**

## The high-voltage system

**In addition to the high-voltage battery, the Volkswagen electric vehicles also have one or more 12 V on-board batteries.**

### Classification as a high-voltage component or high-voltage system is dependent on the voltage type (“AC” or “DC”) in the motor vehicle.

Alternating voltages (AC) above 30 V supply voltage and direct voltages (DC) above 60 V supply voltage are generally referred to as high-voltage components or a high-voltage system.

**Even if the terms are based on the amount of voltage, the actual danger lies in direct contact with electrical power in the rated current with which the closed rated current flows through the human body. This means that, even at a low voltage, contact with electrical power at a correspondingly high current can pose a risk to life.**

**Do not touch, cut or open high-voltage components or the high-voltage battery!**
**Use appropriate personal protective equipment!**

**Only a few electrical components in high-voltage vehicles are powered with high voltage (e.g. High-voltage battery, high-voltage cables, power electronics, electric drive motor/alternator, A/C compressor, external charging socket). All remaining electric components, such as lighting, on-board electronics, etc., are supplied with power via the 12 V vehicle electrical voltage.**

High-voltage batteries are rechargeable batteries. Depending on the manufacturer and vehicle, different battery types are used. They differ in the chemical components used for the battery cells for the anode, cathode and electrolyte and the battery cell design (round, prismatic, pouch).
Currently, lithium-ion batteries (Li-ion) are fitted, for example.
The sizes and fitting locations of the high-voltage batteries vary depending on the vehicle type. A pure electric vehicle requires a larger high-voltage battery than a hybrid vehicle.

The following battery concepts and fitting locations of high-voltage batteries are currently commonly used.
- Under almost the entire vehicle floor
- Under the vehicle floor in front of the rear axle
- Between the axles

A high-voltage battery consists of a wide range of battery modules, which, in turn, consist of the actual battery cells.
All high-voltage batteries are protected by design in order to reduce, for example, the risk of electrolyte leaking from battery cells to a minimum.
In the event of an accident, the high-voltage battery is mechanically protected by battery housing. This largely directs the impact energy into the vehicle structure.

## Warning labels on high-voltage components

Part of the safety concept of high-voltage vehicles, for example, consists of a comprehensive warning label.

**Example of a high-voltage battery from ID.4**

All high-voltage components are marked by clear warning stickers. The high-voltage cables are excluded from this; they catch the eye due to the orange warning color of the cable sheathing.

In principle, three types of warning stickers are used:

- Yellow warning sticker with the warning sign for voltage,
- Warning sticker with the lettering “Danger” on a red background,
- Stickers that particularly warn people with pacemakers.

The yellow stickers indicate high-voltage components that are fitted in the vicinity of the sticker or are hidden below covers.

The warning stickers with the lettering “Danger” directly identify the high-voltage components.

**Examples of warning stickers in high-voltage vehicles:**

# The high-voltage battery

High-voltage batteries are rechargeable batteries. Depending on the manufacturer and vehicle, different battery types are used. They differ in the chemical components used for the battery cells for the anode, cathode and electrolyte and the battery cell design (round, prismatic, pouch).
The high-voltage batteries are lithium-ion batteries. The high-voltage battery is arranged in a stable housing in areas in vehicles that are protected against deformation in most crashes. The sizes and fitting locations of the high-voltage batteries vary depending on the vehicle type. A pure electric vehicle requires a larger high-voltage battery than a hybrid vehicle.
In electric vehicles, the high-voltage battery is usually bolted in the center below the vehicle as a load-bearing vehicle body component. In hybrid vehicles, the high-voltage battery can usually be found in the rear vehicle compartment (in front of behind the rear axle).
In both hybrid and electric vehicles, the high-voltage battery consists of battery cells connected in series which are interconnected in modules. Several modules are installed together with the periphery in a metal housing. The housing is connected to the vehicle via a potential equalization line.
All high-voltage batteries are fitted in a stable housing in order to protect the battery cells during an accident and to prevent electrolyte from leaking from defective battery cells.

**Depending on the vehicle variant, the high-voltage battery may consist of multiple battery packages.**

**In addition to the high-voltage battery, the Volkswagen electric vehicles also have one or more 12 V on-board batteries.**

**Due to the wide range of different battery types with their different chemical components and the constant development of accumulator technology, it is not possible to address their specific hazards and possible behavior in this Repair Manual.**

If the high-voltage battery is damaged or overheated, this can result in exothermic chemical reactions (thermal runaway): These reactions lead to the battery cells quickly heating up. The battery will begin to burn and toxic vapors are exposed.
Important information in this round can be found in chapter 6. In case of fire. Information about how to handle the energy stored in the battery can also be found in chapter 8. Towing/transportation/storage.

## Lithium-ion battery disconnected from the vehicle

If, during an accident, the high-voltage energy storage unit and/or parts of it are disconnected from the vehicle, it is to be assumed that there are electric, chemical, mechanical and thermal hazards from the high-voltage energy storage unit.
In this case, the following points must be taken into account:

**If high-voltage energy storage units, high-voltage components or high-voltage cables are damaged, e.g. open components or torn-off lines, touching these damaged areas must be avoided wherever possible!**

**When working with a hydraulic rescue device when lifting, securing, towing or pulling the vehicle, the location of the high-voltage components and high-voltage cables must be noted (see vehicle-specific rescue card)!**

**In the case of unavoidable work in these areas, damaged parts or high-voltage energy storage units must be covered with electrical insulation. It is recommended to use suitable an electrically insulating flexible cover (undamaged plastic film or another suitable electrically insulating cover, e.g. in accordance with IEC 61112).**

If a high-voltage energy storage unit is disconnected from the vehicle, additional parts of the entire energy storage unit may be in or on the vehicle.
Separated components if high-voltage energy storage units must only be lifted from the ground with electrically insulating equipment!

**In order to protect the face, people should only work with the helmet visor folded down.**

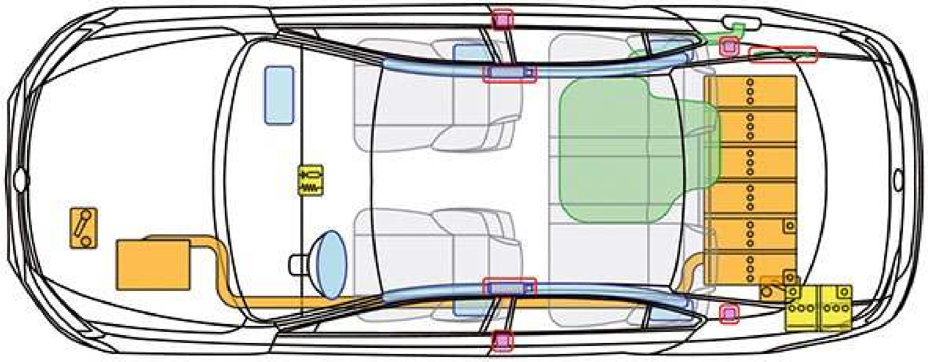
Leaking liquids from high-voltage energy storage units are usually coolants. Electrolytes are only present in low amounts (millimeters) in the individual battery cells.

High-voltage battery vents directed downward

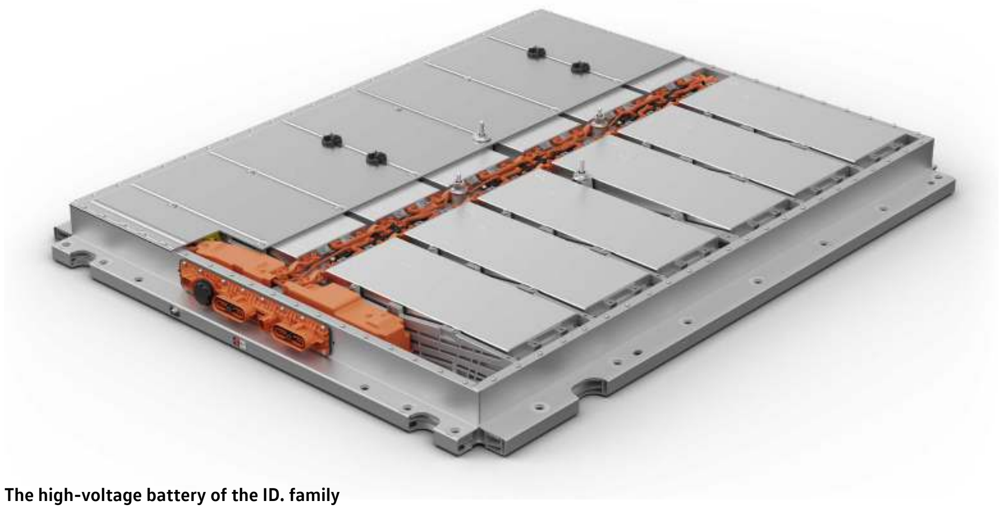
**Electrolytes leaking from damaged high-voltage energy storage units are irritating, flammable and potentially corrosive. Please use appropriate personal protective equipment!**

## Example battery concepts

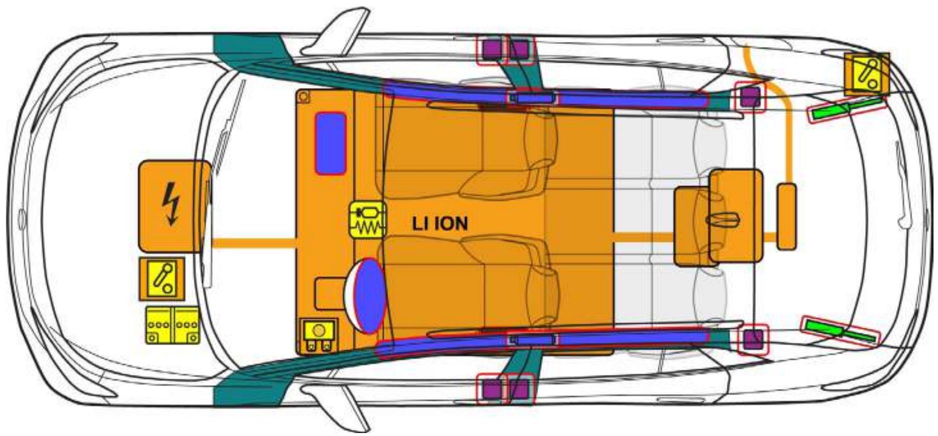
The high-voltage battery of the Jetta hybrid



Installation location of the high-voltage battery in the Jetta hybrid
(The illustration does not correspond to the current ISO:17840)



The high-voltage battery of the ID. family




Installation location of the high-voltage battery in the ID.4

# A/C system

For manual air conditioning (AC), the coolantsR134 a, R1234 yf, R744 ($CO_2$) are used.

# High-voltage battery – cooling system

In normal operating conditions, there is no risk of exposure of the battery to its contents.

**If there is leaking refrigerant from the battery cooling system, there is the risk of a thermal reaction in the high-voltage battery. Monitor the temperature of the high-voltage battery!**

**Toxic fumes can be produced when outgassing the high-voltage battery.**
**Use appropriate personal protective equipment!**

**Vehicle-specific information is also described in the respective rescue cards.**

# 12 V on-board vehicle battery

In Volkswagen models, 12 V vehicle batteries in lead-acid technology are primarily used. The 12 V lead-acid batteries differ with a spill-proof technology (completely black bock and lettering “AGM” on the label) and with a non-spill-proof technology in the event of damage to the housing (which can be identified by a black lid and transparent box). Both technologies use “sulfuric acid” as an electrolyte.

**There may be a highly explosive gas mixture in the battery. No fire, sparks, open flames or smoking in the vicinity of the battery! Use appropriate personal protective equipment!**

**Leaking battery acid can cause serious skin burns.**

**“Explosive” sticker on battery**

**Vehicle-specific information is also described in the respective rescue cards.**

**Batteries with solid electrolyte**

Absorbent glass mat batteries, also known as AGM batteries or recombination-type batteries, are used in vehicles with a start-stop system and recuperation. Absorbent glass mat batteries are batteries in which the sulfuric acid is bound in a fiberglass fleece (AGM). This battery type can be identified by the lettering AGM on the battery cover and the completely black battery housing.

**Leaking battery acid is highly flammable.**

**Vehicle-specific information is also described in the respective rescue cards.**

AIR

## Compressed air tank

In some Volkswagen models, accumulators are fitted for, e.g. air suspension or manual air conditioning (AC). Do not damage these accumulators and never open them with force.

## Flammable materials

These include, e.g.:

- Plastics
- Electrolytes
- Resins
- Magnesium
- Gas or other flammable liquids

Resins are used to connect carbon fibers; magnesium components are found in the engine compartment.

**Avoid skin contact and inhaling electrolyte vapors because electrolytes are flammable, corrosive and irritating.**
**Please use appropriate personal protective equipment!**

**Handling the contaminated extinguishing water is based on the country-specific approach of the rescue and recovery workers.**

# 6. In case of fire

## General information about vehicle fires

In principle, comply with all country-specific regulations, work instructions and guidelines from fire service associations and authorities on how to proceed in the event of a vehicle fire. Where possible, the fire must be prevented from spreading the energy storage unit (fuel, battery).

All commonly used and well-known extinguishing agents such as water, foam, $CO_2$ or powder can be used.
It can only be decided which extinguishing agent and which extinguishing method should be used at the job site and heavily depends on the situation at hand and the equipment available.

**If the airbags have not triggered in an accident, they may trigger in the event of a vehicle fire.**

# Fire in high-voltage vehicles

As a rule, handling high-voltage vehicles is not more dangerous than handling gasoline vehicles; however they do differ in some points. Knowledge of these differences can be crucial during rescue operations for accidents.

During a fire involving high-voltage vehicles, a distinction must be drawn:

- **Vehicle fire without the energy storage unit on fire and with flammable electrolyte:**
  In an identical manner to a conventionally powered vehicle, depending on the requirements and/or availability in a "usual" fire of s a hybrid or electric vehicle (HEV or BEV, where the high-voltage battery is not on fire), all commonly used and well-known extinguishing agents, such as water, foam, $CO_2$ or powder can be used.
- **Vehicle fire with the energy storage unit on fire with flammable electrolyte:**
  Smoke, flying sparks and flash flames from the battery may indicate that the lithium-ion battery is contributing to the fire.
  If a high-voltage battery is on fire, you should extinguish it with water where possible and then cool it down.
  In this case, it must be ensured that sufficient water is used and, where possible, extinguishing water enters into the high-voltage battery through openings created by fire or collisions.
  The water jet should be directed as directly as possible at the battery.
  The installation position of the high-voltage battery can be found on the rescue card for the relevant model.

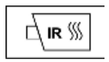
The decision about appropriate measures is made at the job site by the fire services and is heavily dependent on the situation at hand (e.g. how the fire is developing and the time of arrival of the fire service) and the equipment available.

**Is the electric vehicle on fire?**
→ No: **Use rescue cards to immobilize the vehicle, to disable the high-voltage system and to disconnect the 12 V starter battery.**
↓ Yes

**Is the high-voltage battery affected by fire?**
→ No: **Extinguish in the conventional way. Prevent extinguishing water from entering the high-voltage battery.**
↓ Yes

**Extinguish with sufficient water and cool, if possible with extinguishing water entering the high-voltage battery!**

**Flowchart of fire in electrified vehicles**

| | | |
|---|---|---|
|  | **If there is a thermal reaction in the high-voltage battery, hot, poisonous gases may escape.** | |
|  | **Use large amounts of water** |  |
|  | **In the event of damage or improper use, lithium-ion batteries may even ignite promptly or with a delay or even re-ignite after a fire has been extinguished. Use appropriate personal protective equipment!** |  |

A lithium-ion battery may react promptly or with a delay due to severe damage (e.g. pressed, broken or cracked housing), exposure to water or fire. Therefore, while carrying out operations on a vehicle involved in an accident with a lithium-ion battery, pay attention to signs of a reaction (e.g. smoke, heating, noises, sparks, etc.).

In the event of reaction of the lithium-ion battery, protective measures and counter measures must be taken.
As in conventionally powered vehicles, fires in electric vehicles result in fire smoke that is detrimental to health. Therefore, it is recommended to use appropriate protective equipment.
In the event of a fire, outgassing of the high-voltage battery must be expected. The battery has mechanical safety devices (also known as vents) which open during a rise in temperature or pressure caused by a fire, for example, and therefore lead to a targeted "outgassing" and depressurization.

**High-voltage battery vents directed downward**

It is possible to extinguish a vehicle with a high-voltage battery and to extinguish a high-voltage battery that is on fire. Water is preferable as a extinguishing agent and is not fundamentally different from fighting fires in a conventionally powered vehicle.

If the high-voltage battery is contributing to a fire, larger amounts of extinguishing water are required or extinguishing an undamaged, reacting high-voltage battery.

After a reaction, the lithium-ion battery must be cooled with water until it has reached approximately the ambient temperature. Use of a thermal imaging camera or an IR thermometer is recommended.

**After fighting a fire, dangerous electrical voltages may still be present.**

**If batteries are not completely burned out, there is the possibility of re-ignited. Extinguished vehicles must be parked at an appropriate storage location (quarantine spot); if necessary, the vehicle must be observed.**

**A sufficient safety distance must be maintained. Corresponding self-contained breathing equipment must be worn!**

**Vapors and gases can be suppressed with a water spray.**

**Bursting of exposed defective battery cells with an associated exothermic reaction cannot be excluded.**

**It is possible that there may be a fire at a later time after the accident because the residual risk of a delayed emergence of a fire cannot be excluded. This particularly applies in the case of damaged high-voltage energy storage units (see also chapter 8. Towing/transportation/storage). Electrical hazards are also still possible. High-voltage components must not be touched and it must be ensured that appropriate personal protective equipment is worn. High-voltage cables may have been damaged by the heat.**

## 6. In case of fire

**Further information can be found in the respective rescue cards.**

# 7. In case of submersion

## Vehicle under water

A vehicle immersed in water must be treated in the same way as a damaged vehicle involved in an accident.
The safety and security regulations must be complied with and the process for eliminating the direct hazards must be followed, see section 3. Disable direct hazards/ safety regulations.

## High-voltage vehicle under water

- There is fundamentally no increased risk of electric shock in water from the high-voltage system.
- The same instructions as in chapter 3. Disable direct hazards/safety regulations apply.
- The recovery process is identical to that for conventional vehicles.

**If water enters the high-voltage battery, an electrolysis can be started, which can lead to a deflagration of oxyhydrogen.**

**The high-voltage system must be disabled (see chapter 3. Disable direct hazards/safety regulations).**
**Use appropriate personal protective equipment!**

**In the event of heavily soaked vehicles, Disconnecting the 12 V vehicle battery is recommended due to the risk from an electrolysis.**

# 8. Towing/transportation/storage

# Recovering vehicles involved in accidents

The information on the rescue cards must be followed during loading, transportation and storage.

Example illustration (front towing eye)

Example illustration (rear towing eye)

**If a trailer hitch is installed, it is not possible to screw in a towing eye at the rear of the vehicle. Use the ball head of the trailer hitch.**

**Uncouple trailers connected to the vehicle before transport and transport separately.**

**In some Volkswagen models, the trailer hitch is unlocked electrically. See also Disconnecting the 12 V vehicle battery.**

**The rescue and emergency services on-site decide about the relevant approach.**

# Recovery high-voltage vehicles involved in accidents from a hazardous area

In principle, vehicles with high-voltage batteries should be taken away on platform vehicles.

Before transportation, the high-voltage system must be disable, see chapter 3. Disable direct hazards/safety regulations.

When transporting the vehicle away (e.g. by a towing company), the state of the lithium-ion high-voltage battery must be checked again. The vehicle can only be loaded and taken away if the vehicle does not show any signs of a reaction in the area of the lithium-ion high-voltage battery over an extended period of time, see flowchart on the next page.

For vehicles involved in an accident with a damaged or conspicuous lithium-ion battery, you must wait for the resolution of the reaction of the lithium-ion battery such that no further reaction on the transport route is expected, see flowchart on the next page. The shortest possible and least dangerous route must be selected. Driving through tunnels should be avoided.

If necessary or in the case of doubt, it may be necessary to have the tow truck accompanied by a fire truck.
Vehicles with a damaged high-voltage battery should be transported to a safe storage location.

After transportation, electric and hybrid vehicles involved in accident should be parked outdoors and not in enclosed buildings at a sufficient distance from other vehicles, buildings, flammable objects or flammable surfaces.

Designated “quarantine surfaces” at the storage location should be preferably used. Due to the theoretically still existing reaction possibilities of the lithium-ion battery, the vehicle involved in an accident must be parked outdoors at a suitable place. The parking area must be identified accordingly (signage and boundaries).
A minimum distance of 50 ft must be maintained from other vehicles, buildings or flammable objects. The distance can be reduced by taking corresponding measures, e.g. such as a firewall. The regional specifications must be followed.
The responsible persons at the towing company, qualified workshops and, if necessary, the disposal companies must be informed of the special characteristics and risks of the vehicle!

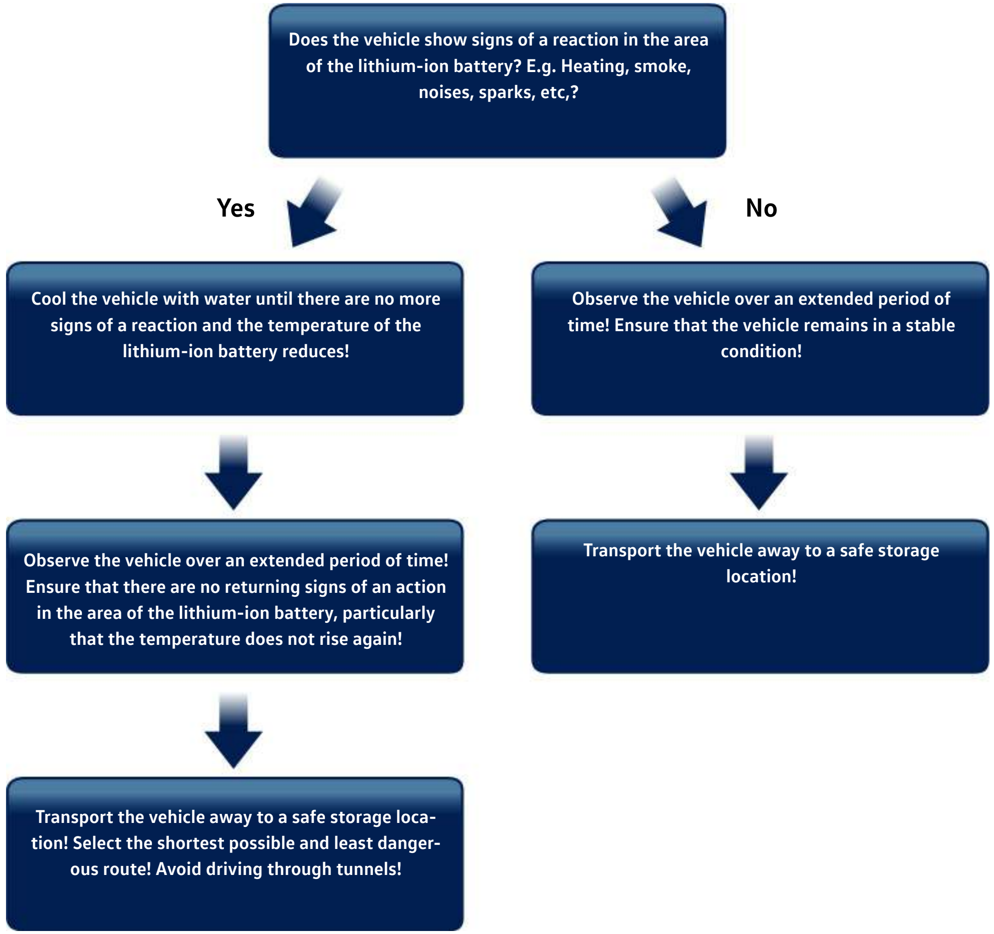
**Lithium-ion batteries can self-ignite or reignite after firefighting measures have been taken!**

**If the vehicle has been involved in an accident or if the high-voltage battery is damaged or not working properly: Disable the high-voltage system (see chapter 3. Disable direct hazards/safety regulations). Park the vehicle a safe distance of at least 50 feet away from buildings and other vehicles (quarantine surface).**

**During loading, ensure that the high-voltage components are not damaged. If possible, lift the vehicle by the marked lifting points.**

**High-voltage batteries may self-ignite again during transportation due to vibrations.**

| |
|---|
| **Vehicle-specific recommendations are described in the respective rescue cards.** |
| **Monitor the temperature development, ideally with corresponding devices, e,g, IR camera, over an extended period of time!** |
| **A larger metal container is recommended for transporting away a high-voltage energy storage unit or parts of it that have been separated from the vehicle. The condition of the high-voltage energy storage unit must be observed (e.g. smoke development, noises, sparks, heat development) and a flooding of the metal container must be prepared for.** |
| **For further information about this, see chapter 5. Stored energy/liquids/gases/solids (lithium-ion battery disconnected from the vehicle).** |




Flow chart for towing electric vehicles

# 9. Important additional information

Depending on the vehicle type and equipment version, current motor vehicle may have extensive vehicle occupant protection systems.

## Airbag

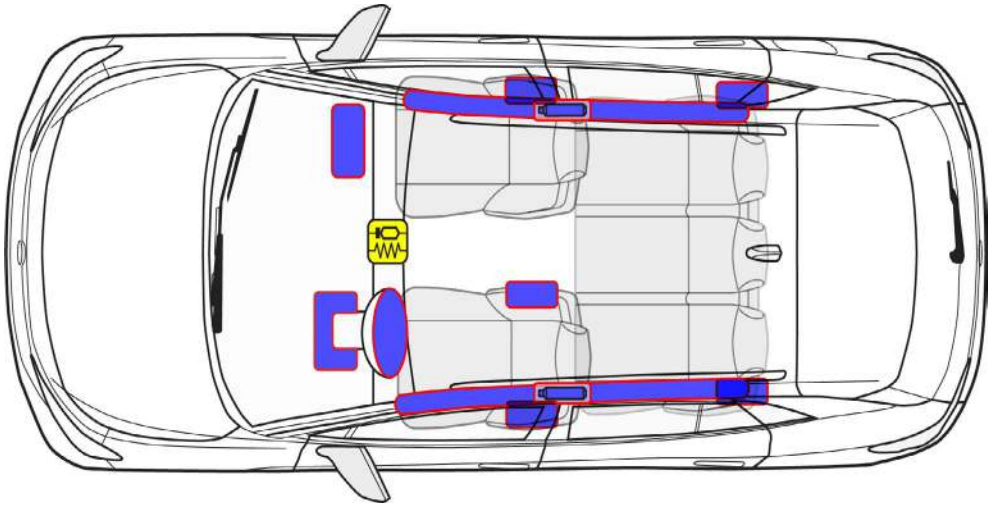
A current and maximally equipped vehicle includes the main components:

- Airbags
- Airbag control unit
- Sensors
- Seat belt pretensioners and
- In Cabriolets, the triggering components for the rollover bar

They are triggered via pre-tensioned spring or pyrotechnically. The electronics integrated in the airbag control unit has the task to record the vehicle deceleration or vehicle acceleration and to detect whether it is necessary to trigger protective systems. In addition to the sensors in the airbag control unit, sensors in the front doors, for example, are also used to record the vehicle deceleration or vehicle acceleration during an accident. Only once the information from all sensors is evaluated will the electronics in the airbag control unit decide whether or when which safety components are activated. Depending on the type and severity of the accident, only the seat belt pretensioners or the seat belt pretensioners together with the airbags are triggered.

The control module is identified in the rescue cards as follows:

**Airbag control unit**



**Airbags in contemporary vehicle models**

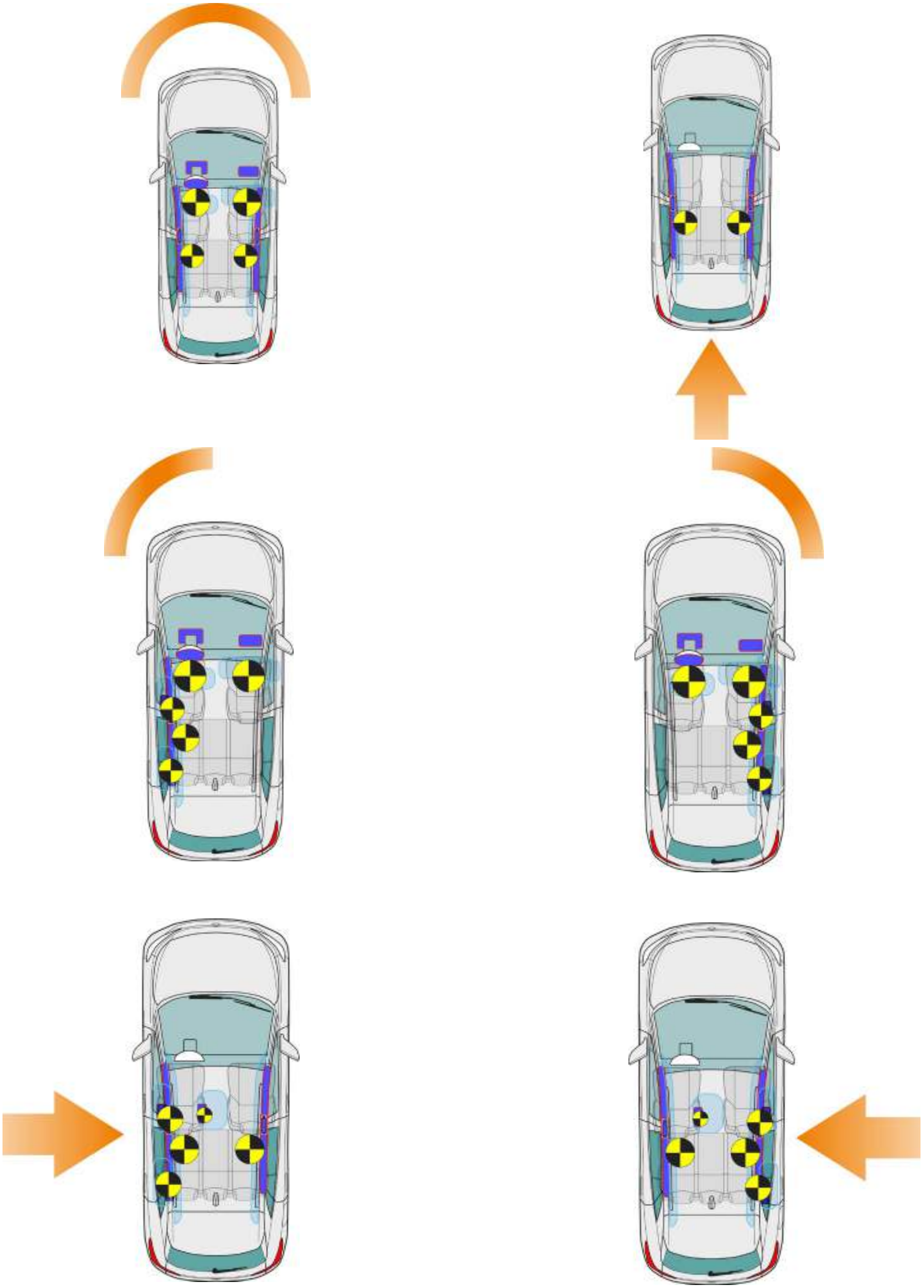
Only those safety systems are triggered that have a protective function in the specific accident situation.

In addition to the main function of controlling the airbag, the airbag control unit may have the following additional functions:

- Emergency release of the central locking system
- Switching on the interior lighting
- Switching off the fuel pump
- Switching on the emergency flashers
- Transferring a signal for sending the emergency call service
- Opening the windows after an accident
- Switching off the A/C system

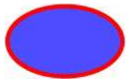
Inflators generate the amount of gas required to fill the airbag and therefore blow up the airbags within milliseconds. The inflated airbags protect the seatbelted vehicle occupants in a serious accident prior to an impact on internal body contours (e.g. the steering wheel, the instrument panel, etc.)
Depending on the installation location and requirement, inflators of different construction shapes or with different operating principles are used.

**The safety systems are triggered depending on the type of accident or the direction of the impact.**



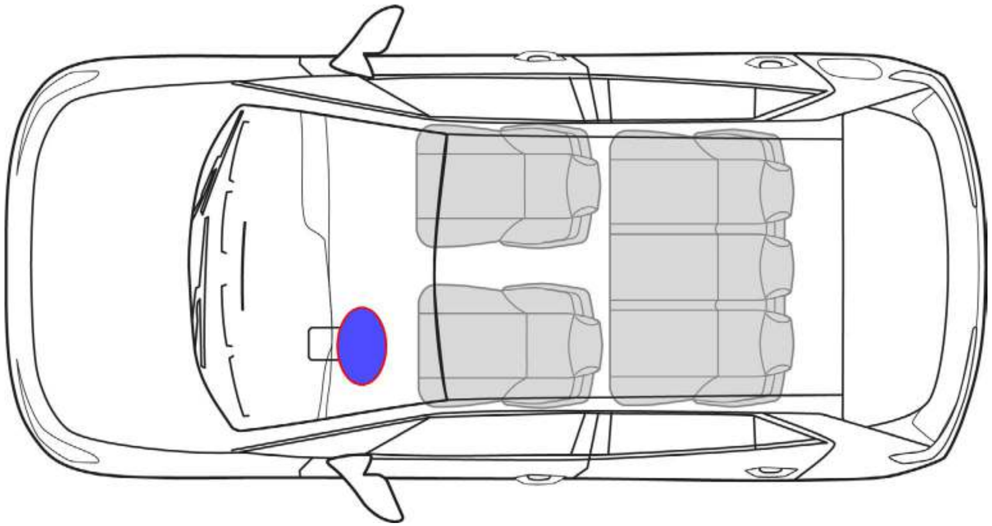
**The safety systems are triggered depending on the type of accident or the direction of the impact (ms = milliseconds).**


Airbags are identified in the rescue cards as a symbol or the contour is identified accordingly as follows:

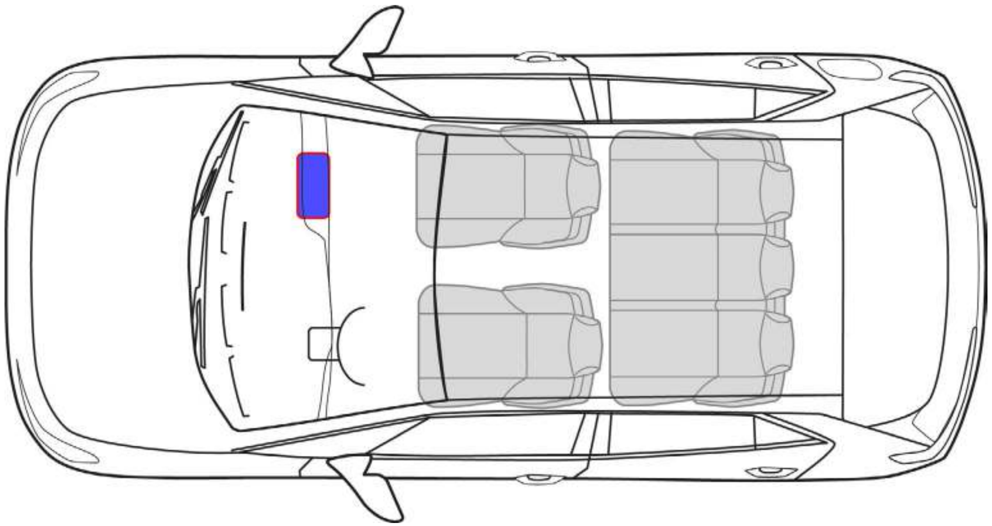
| | |
|---|---|
|  | **Driver's airbag, front passenger's front airbag, side or central airbag, knee airbag and head curtain airbag** |

## Front airbag

**Driver's airbag**
The driver airbag unit essentially consists of a cap, airbag and inflator. It is secured in the steering wheel and is electrically connected to the airbag control unit via a contact unit.
The airbag is folded together below the cap and is designed with respect to its shape and size such that it is protectively erected between the driver and steering wheel after filling.
An inflator blows up the driver's airbag. The unloading airbag opens the cap of the steering wheel at a predetermined score line and is filled with gas in the shortest possible time. The entire process for igniting the inflator until the airbag is inflated takes a few milliseconds.
The kinetic energy is transferred via ventilation openings facing away from the driver when immersing the upper body by evenly venting the filling gas.

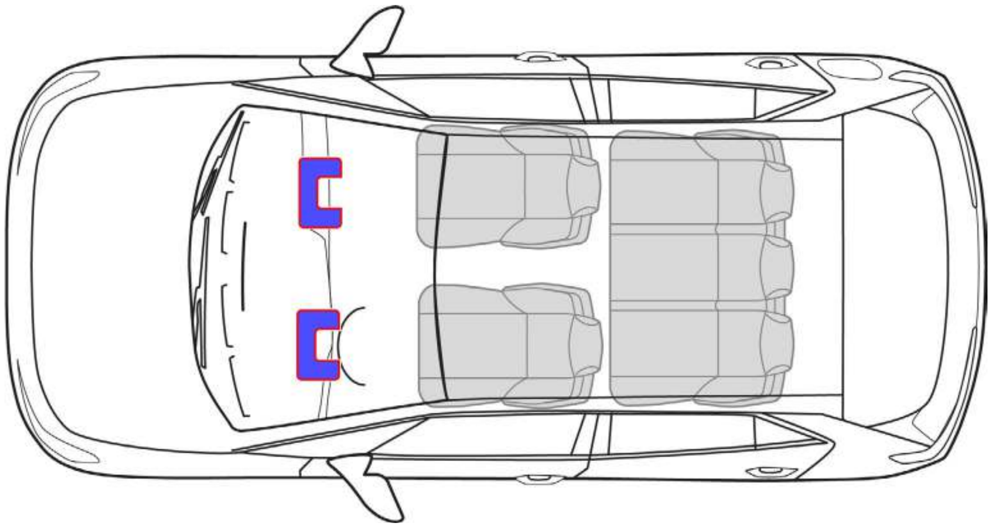


**Front passenger's front airbag**
The airbag unit for the front passenger is located in the instrument panel in front of the front passenger seat.
Due to the larger distance between the airbag unit and the vehicle occupant, the airbag of the front passenger's front airbag has a much larger volume. The function of the front passenger's front airbag, how it operates and its timing are comparable with those of the driver's airbag.



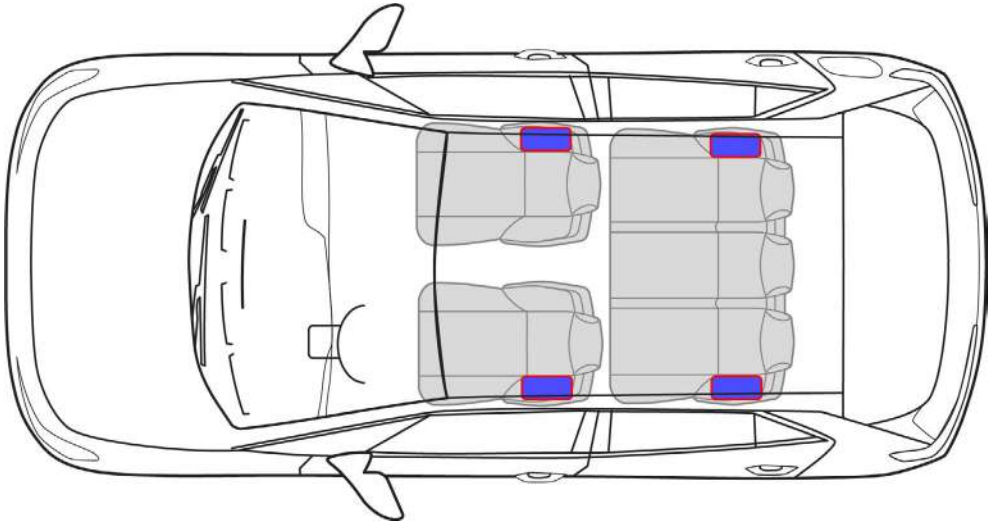
### Knee airbag

The design of the knee airbag is comparable to the design of the front passenger's front airbag. It is located in the footwell trim below the instrument panel.
The knee airbag is always triggered together with the driver's airbag. Single-stage inflators are used to inflate the knee airbag.
The triggered knee airbag reduces the potential for injury of vehicle occupants in the knee and leg area and the vehicle occupant is coupled to the vehicle decelerations at an earlier stage.



## Side airbags

During accidents from the side, side airbags protect the thorax and back of vehicle occupants on the vehicle side facing the impact and reduce the load. The inflate at the side between the upper body and intruding trim parts and therefore distribute the loads more evenly on the vehicle occupants, which is thus coupled to the intrusion movement at an early stage.
The side airbags are located in the backrest of the driver and front passenger seat and, in some Volkswagen models, at the outer seats of the second row of seats. As a result, this ensures a consistent distance from the vehicle occupant in every seat position.



## Head/thorax airbag

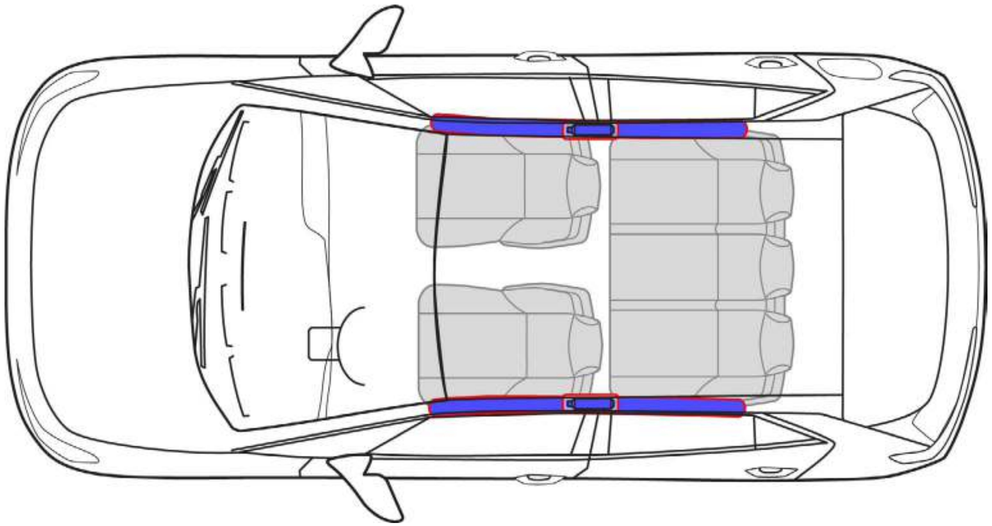
The head/thorax airbag for the driver and front passenger are both integrated in the backrests of the front seats. The design and function are comparable to those of a side airbag. It extends across the vehicle occupant’s rib cage to their head and is specifically installed in Cabriolets, where a head curtain airbag is not possible.

## Head curtain airbag

Head curtain airbags are used to protect the head in the event of a side impact. The consist of a large-area airbag that usually extends at the top of the vehicle lining from the A-pillar to the C-pillar.

Depending on the vehicle model, the inflators in the roof area may be fitted to the B-pillar or between the B-pillar and C-pillar or between the C-pillar and D-pillar or at the back of the roof area. The precise installation position is described in the rescue cards.
Unlike front and side airbags, the head curtain airbag may retain its internal pressure for some time after triggering, in order to also provide a protective effect in the event of subsequent vehicle rollovers or secondary collisions.

Both side and head curtain airbags are triggered by the airbag control unit if a limit value stored there is reached. A side impact is recorded by lateral acceleration sensors or pressure sensors in the doors.



# Air bag inflators

## Solid fuel inflators

Solid fuel inflators consist of a housing in which a solid fuel charge with an ignition unit is integrated. After the solid fuel is ignited, filling gas that is harmless to the vehicle occupants is produced.

**Procedure:**

- The igniter is activated by the airbag control unit.
- The propellant charged is ignited and suddenly burns.
- The produced gas flows through the metal filter into the airbag.

## Hybrid inflators

The hybrid inflators consists of a housing in which a compressed gas stored under high pressure and a solid fuel charge with an ignition unit are combined. The design and shape of the generator housing are adapted to the respective installation conditions. These inflators are generally tubular. The main components are the pressure vessel with the airbag filling gas and the propellant charge integrated in the pressure vessel or flange-mounted against it (solid fuel). The solid fuel is used in tablets or in a ring shape. The stored and compressed gas is a mixture of noble gases, e.g. argon and helium. Depending on the inflators' design, it is under a pressure of between 2900 psi and 11,603 psi.

Igniting the solid fuel opens the pressure vessel and produces a gas mixture from the gas of the solid propellant charge and the noble gas mixture. The igniter is activated by the airbag control unit and the propellant charge is ignited.

**Do not damage inflators during rescue work. The compressed gas in the pressure vessel and the pyrotechnic fuels may pose a potential danger to rescue workers and vehicle occupants.**

# Safety belt pretensioners

Seat belt pretensioners wind the seat belt against the pull direction in the event of a crash, such that the slack (space between the strap and body) is reduced. This prevents vehicle occupants from making a forward movement (relative to the vehicle movement) at an early stage, A seat belt pretensioner is able to roll up to approx. 8 inches within approx. 10 milliseconds. The seat belt pretensioners are integrated within the strap system. However, depending on the vehicle type, they may installed in different spaces (e.g. in the B-pillar, in the side member next to the seat or on the outer side of the rear seat) and have different functional principles. If necessary, two seat belt pretensioners may be used on a single seat.

**Therefore, seat belt pretensioners should not be damaged with rescue equipment where possible. You must avoid striking this area!**

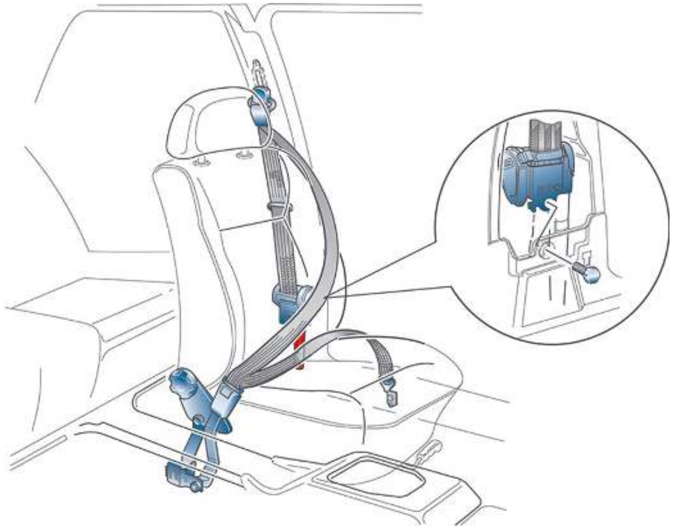
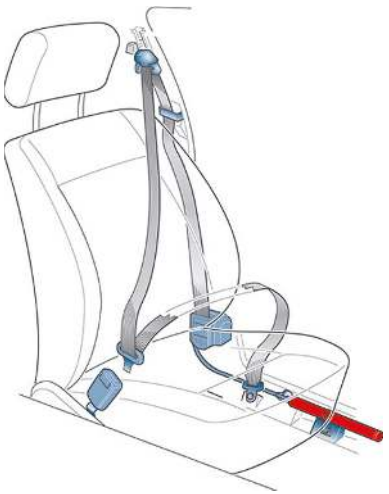
**The seat belt also locks if the vehicle is strongly inclined, upside down or if the seat belt pretensioners has been damaged by the accident.**

**Seat belt pretensioners with mechanical triggering that have not been triggered may still trigger after the battery has been disconnected.**

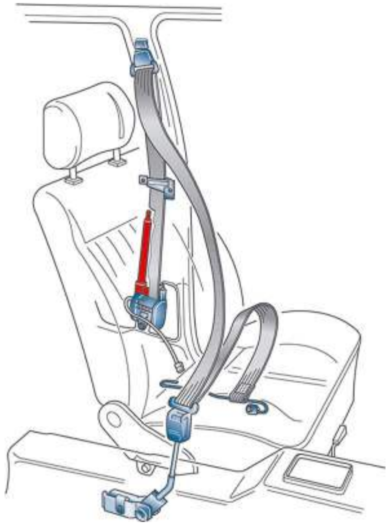
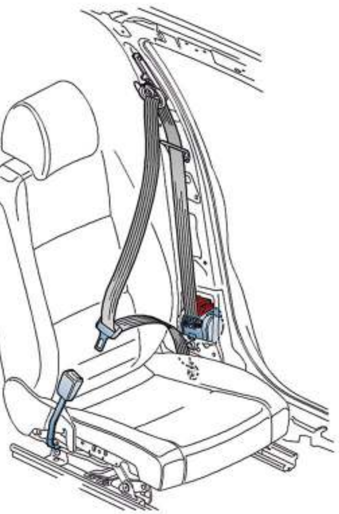
**The safety belt should, if the situation allows, be loosened or cut off as soon as possible.**

**Safety belt pretensioner**

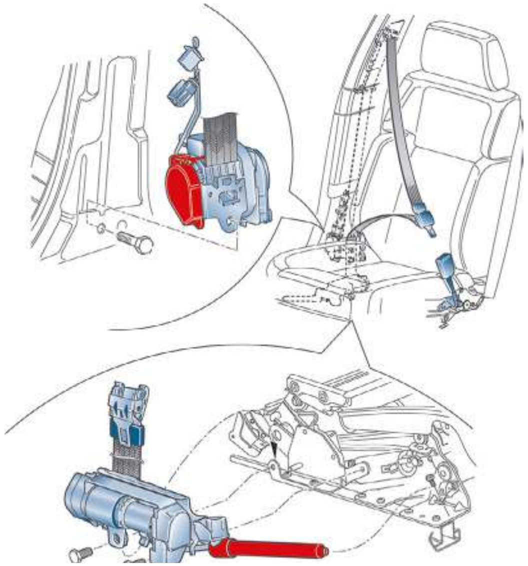
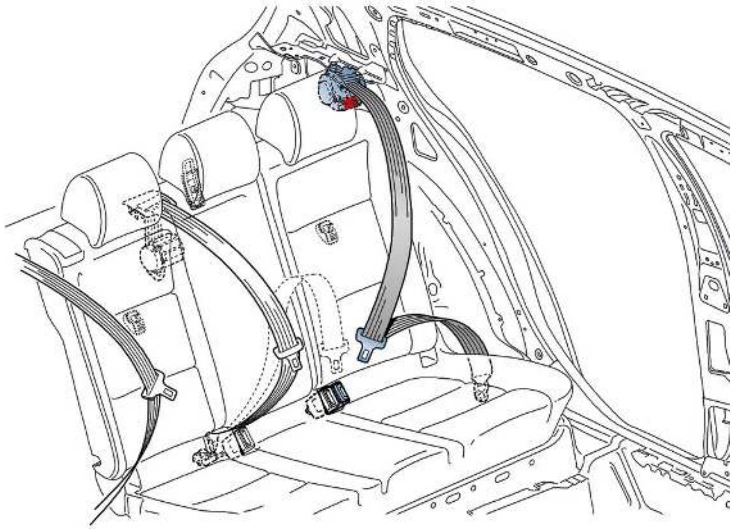
## Installation variants for the safety belt pretensioner

| Version | Component location |
|---|---|
|  | **Version 1**<br>The three-point seat belt with the cylinder-shaped seat belt pretensioner and the mechanical or<br><br>electrical triggering of the ignition form one unit and are installed either:<br><br>a) In the B-pillar below the automatic belt retractor,<br>b) As external components next to the side member or<br>c) In the B-pillar above the automatic belt retractor.<br><br>**Installation variant 1a – safety belt pretensioner in the B-pillar below the automatic belt retractor** |
|  | **Installation variant 1b – safety belt pretensioner as external components next to the side member** |

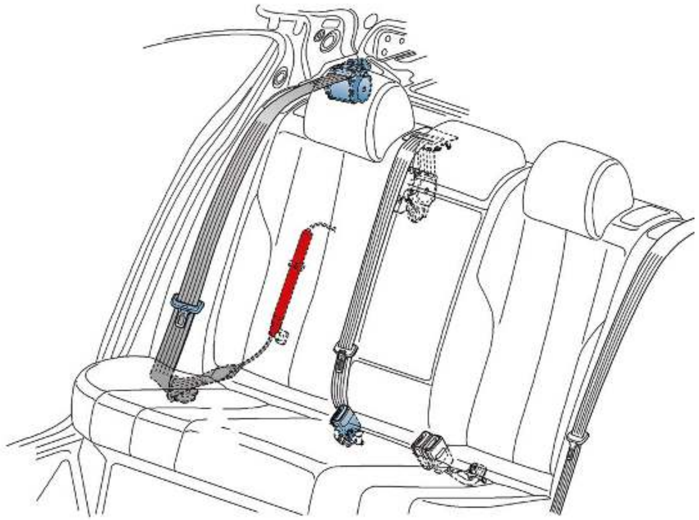
## Installation variants for the safety belt pretensioner

| Version | Component location |
|---|---|
|  | **Installation variant 1c– safety belt pretensioner in the B-pillar above the automatic belt retractor.** |
|  | **Version 2**<br>In the front compact pretensioners, the three-point seat belt and seat belt pretensioner with electrical or mechanical triggering of the ignition form one unit and are installed in the B-pillar.<br><br>**Installation variant 2 – compact belt pretensioner in the B-pillar** |

## Installation variants for the safety belt pretensioner

| Version | Component location |
|---|---|
|  | **Version 3**<br>In the front double pretensioner, the shoulder section of the belt with a compact belt pretensioner and the lap section of the seat belt with a cylinder-shaped seat belt pretensioner form a functional unit.<br><br>In the shoulder section of the belt, the electric triggering of the ignition is installed in the B-pillar, and in the lap section of the seat belt it is installed on the seat frame.<br><br>**Installation variant 3 – double belt pretensioner in the B-pillar and seat frame** |
|  | **Version 4**<br>In the back compact pretensioners, the three-point seat belt and seat belt pretensioner with electrical or mechanical triggering of the ignition form one unit and are installed behind the rear seat backrest.<br><br>**Installation variant 4 – behind compact belt pretensioner in the rear shelf** |

## Installation variants for the safety belt pretensioner

| Version | Component location |
|---|---|
|  | **Version 5**<br>The three-point safety belt and the seat belt pretensioner are arranged separately from each other. The seat belt pretensioner with electrical triggering of the ignition is installed in the area of the wheel housing/C-pillar.<br><br>**Installation variant 5 – rear seat belt pretensioner in the area of the wheel housing/C-pillar** |
|  | **Version 6**<br>The three-point safety belt and lap belt tensioner are fitted separately from each other. The lap belt tensioner with electrical triggering of the ignition is installed on the side member/B-pillar.<br><br>**Installation variant 6 – lap belt tensioner in the area of the side member/B-pillar**<br><br>**Not shown**<br>In the ID. Buzz, the three-point safety belt tensioner and the lap belt tensioner (double tensioning) are installed together in the B-pillar. |

# Automatic Rollover Support System

Cabriolets must offer vehicle occupants the maximum possible protection even when the roof is open. Therefore, a rollover support system is used, which, in combination with reinforced A-pillars, creates a protection zone for vehicle occupants. This can be rigid or dynamic.

The following functionality applies to a dynamic system:

- In the airbag control unit, there is a sensor for detecting the risk of a rollover.

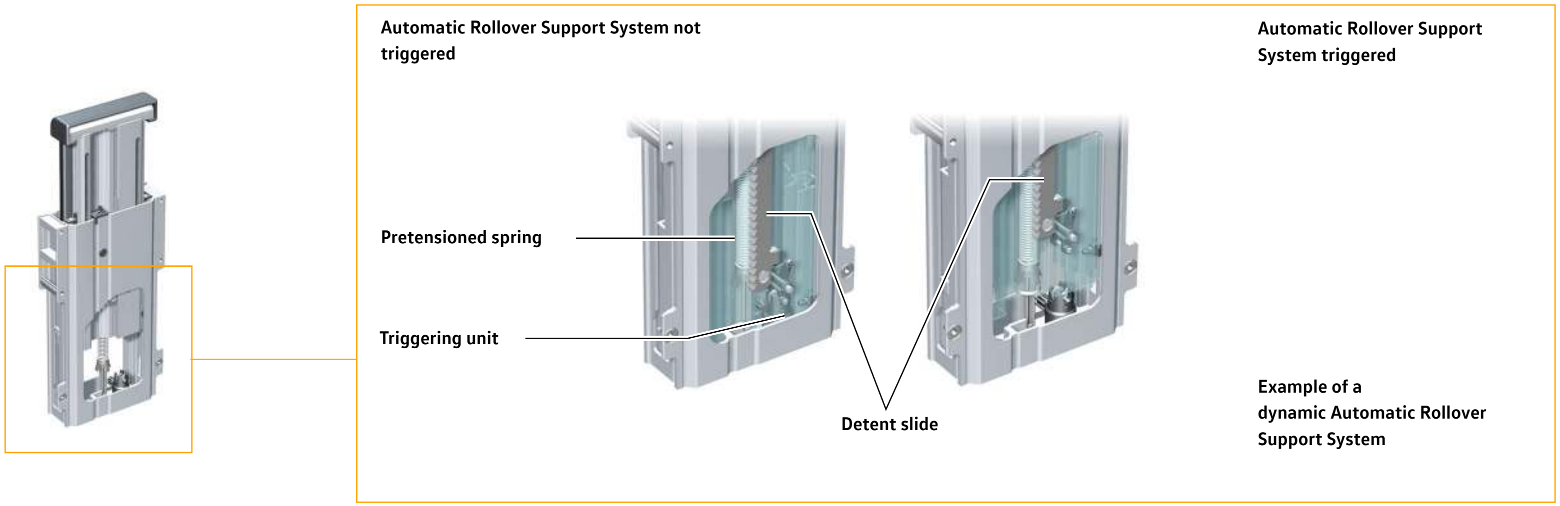
Together with other sensors fitted in the control modules, the severity of the accident is determined and the Automatic Rollover Support System and the seat belt pretensioners are triggered.

The Automatic Rollover Support System is also triggered as a precautionary measure in the event of a front, side or rear impact with a high accident severity as soon as a seat belt pretensioner or airbag is ignited.

It is triggered by a triggering unit of the Automatic Rollover Support System. Using a pretensioned spring, the bracket is moved to the protection position within approx. 0.25 seconds and locked in the extended state with the detent slide.

**If the rear window is still intact when the Automatic Rollover Support System, this is not broken through by the Automatic Rollover Support System. If the pane is removed as part of the rescue measures, the rollover bar will be pressed upward by a further 4 inches. This could hit rescue or recovery workers and hurl around glass splinters.**

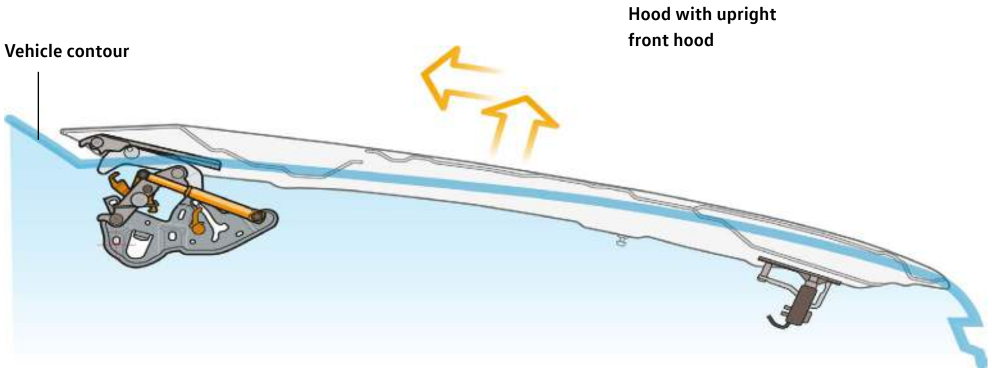
**Labeling of Automatic Rollover Support System**




Example of a dynamic Automatic Rollover Support System

# Re-active hood

In order to ensure optimal pedestrian protection, some Volkswagen vehicle models are equipped with an active hood.
The re-active hood is lifted in the front and rear areas by pretensioned gas-pressure struts and pyrotechnic fuels in the event of a collision with a pedestrian.
This increases the distance between the hood and engine. In this position, the hood can absorb more impact energy and therefore reduces the severity of injuries from the engine.




Example of re-active hood with pyrotechnic actuator

**Do not damage inflators during rescue work. The compressed gas in the pressure vessel and the pyrotechnic fuels may pose a potential danger to rescue workers and vehicle occupants.**

**Active Pedestrian Protection System**

# 10. Explanation of pictograms used

## Components/function/measures that must be taken into consideration during a rescue operation are shown by special pictograms.

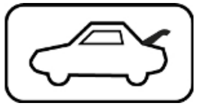
Pictograms:

- Show, together with the rescue card illustration, where the relevant components/ functions are located in the vehicle (for details, see ISO 17840-1 and ISO 17840-2)
- Indicate a certain function or dangers; these can be found in the chapters on the additional pages of the rescue card or the chapters of the Repair Manual for rescue workers
- Determine how to identify the drive type and
- Show fire extinguishing measures.

**Some pictograms can be adapted in this way such that the reflect the actual size and shape.**
**It may also use a combination of simple shapes.**

| Pictograms for detecting the drive type | |
|---|---|
|  | Vehicle with class 2 liquid fuels; gasoline |
|  | Electric vehicle |

## Pictograms for accessing components

| | |
|---|---|
|  | Hood |
|  | Luggage compartment |

## Pictograms for disabling a vehicle (without a high-voltage system)

| | |
|---|---|
|  | Cutting the power to the vehicle |
|  | Remove the digital key |

## Pictograms for disabling a vehicle's high-voltage system (EV, HEV, PHEV, FCEV)

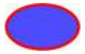
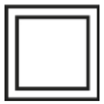
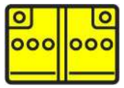
| | |
|---|---|
|  | Dangerous voltage |

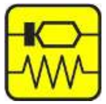
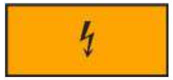
## Pictograms for disabling a vehicle's high-voltage system (EV, HEV, PHEV, FCEV)

| | |
|---|---|
|  | Fuse for switching off the high voltage |
|  | Cable coupling point |
|  | High-voltage disconnection at low-voltage coupling point |

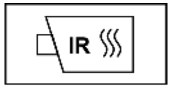
## Pictograms for accessing vehicle occupants

| | |
|---|---|
|  | Adjusting the tilt of the steering wheel |
|  | Seat height adjustment |
|  | Horizontal seat adjustment |

| Other vehicle-related pictograms | |
| --- | --- |
|  | Airbag |
|  | Gas generator |
|  | Safety belt pretensioner |
|  | Gas-pressure strut/preloaded spring |
|  | Active Pedestrian Protection System |
|  | Vehicle body reinforcement |
|  | Area requires particular attention |
|  | Low-voltage battery |

| Other vehicle-related pictograms | |
| --- | --- |
|  | SRS control module |
|  | High-voltage battery |
|  | High-voltage component |
|  | High-voltage cable |
|  | Gasoline/ethanol fuel tank |
|  | Air tank |
|  | A/C system |

## Pictograms for fighting fires and safety

| | |
|---|---|
|  | Danger |
|  | Electrical hazard |
|  | Use IR thermal imaging camera |
|  | Extinguish with water |

## Globally harmonized symbols

| | |
|---|---|
|  | Explosive |
|  | Flammable |
|  | Pressurized gas |

## Globally harmonized symbols

| | |
|---|---|
|  | Corrosive |
|  | Harmful to health |
|  | Dangerous to the environment |